# Phase Linking for Mining Areas: Integrating Deep Neural Networks With Multipolarimetric SAR Data

Ningjie Liu, Bingqian Chen, *Member, IEEE*, Hanwen Yu, *Fellow, IEEE*, Wen Xiao, *Member, IEEE*, Jiahui Liu, and Junyun Wu

***Abstract*—Mining areas are often vegetated and dominated by distributed scatterers (DSs), whose interferometric phases are sensitive to spatiotemporal baselines, causing low coherence and poor phase quality. Phase linking (PL) has emerged as an effective approach for reconstructing consistent phases of DSs, but most methods assume phase stationarity within the estimation window. In mining scenes, large, rapidly varying subsidence breaks this assumption and causes severe phase bias. In addition, many PL methods underuse polarimetric information and rely on biased sample coherence matrix (SCM) estimates, which further degrade PL accuracy. To address these issues, a mining-oriented neural-polarimetric PL (MNP-PL) approach is proposed, which integrates deep learning with multipolarimetric synthetic aperture radar (SAR) data to improve the accuracy of DS phase estimation in mining areas. A deep neural network is first employed to estimate deformation phases and correct nonstationary interferometric phases in mining areas through defringing processing. Then, polarimetric information from the SCM is fused with spatial features to suppress noise and obtain a more accurate SCM. Finally, adaptive nonlinear shrinkage (ANS) is applied to the SCM to further improve the accuracy of the PL. Simulations show that MNP-PL mitigates the effect of dense fringes and SCM-induced noise, yielding more accurate PL estimates. Real data experiments recover more coherent DS phase time series and clear deformation fringes. Compared with the eigendecomposition-based maximum likelihood estimator of the interferometric phase (EMI), total power stacking-based EMI (TP-EMI), MLE-multipolarization PL (MPPL), and TSO-PL, MNP-PL improves deformation estimation accuracy by 68.02%, 63.40%, 60.36%, and 19.30%, respectively. These results show that MNP-PL is accurate and robust for phase reconstruction in low-coherence, densely fringed mining scenes, enabling precise surface deformation mapping.**

***Index Terms*—Distributed scatterers (DSs), mining area deformation monitoring, multipolarized data, phase linking (PL), time-series interferometric synthetic aperture radar.**

## I. Introduction

DIFFERENTIAL interferometric synthetic aperture radar (D-InSAR) systems provide high-precision deformation measurements over large areas of the Earth's surface and are widely used to monitor urban [1], mining [2], landslide [3], and earthquake-related deformations [4]. However, conventional D-InSAR techniques are vulnerable to spatiotemporal decorrelation [5] and atmospheric phase disturbances [6], which can compromise the accuracy of time-series deformation monitoring. To address these limitations, several time-series InSAR (TS-InSAR) methods have been developed, including persistent scatterer InSAR (PS-InSAR) [7], [8], interferometric point target analysis (IPTA) [9], and the Stanford Method for Persistent Scatterers (StaMPS) [10], [11]. Nonetheless, these TS-InSAR approaches, which rely on high-coherence targets, often encounter difficulties in low-coherence regions such as mining areas, where bare soil or sparse vegetation predominates. In these environments, high-coherence point targets are scarce, limiting the number of reliable monitoring points and hindering comprehensive characterization of ground deformation over time [12].

Environmental conditions in mining areas often result in a high prevalence of distributed scatterers (DSs) on the ground surface. Unlike high-coherence point targets, DSs typically exhibit low interferometric phase quality [13]. To address this, researchers have developed various methods to optimize the DS phase quality and increase the density of monitoring points. The main approaches for DS phase optimization include spatial adaptive filtering based on homogeneous pixels [14], [15], [16] and phase linking (PL) techniques [17], [18], [19], [20]. Spatial adaptive filtering methods primarily exploit spatial correlations between pixels but overlook temporal phase correlations. While these methods can effectively reduce phase noise, they do not ensure phase consistency across time-series data. In contrast, PL methods account for both spatial and temporal correlations of DS phases. By employing probabilistic models and applying the phase closure principle

Received 8 September 2025; revised 20 December 2025; accepted 23 December 2025. Date of publication 29 December 2025; date of current version 13 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42374020 and Grant 42271456; in part by the Open Fund of National Engineering Research Center of Geographic Information System, China University of Geosciences, Wuhan, China, under Grant NERCGIS-202403; and in part by the Fundamental Research Funds for the Central Universities, Chang'an University (CHD), under Grant 300102263510. *(Corresponding authors: Bingqian Chen; Hanwen Yu.)*

Ningjie Liu, Jiahui Liu, and Junyun Wu are with the School of Geography, Geomatics and Planning, Jiangsu Normal University, Xuzhou 221116, China (e-mail: liuningjie@jsnu.edu.cn; 2020241664@jsnu.edu.cn; 2020241689@jsnu.edu.cn).

Bingqian Chen is with the School of Geography, Geomatics and Planning, Jiangsu Normal University, Xuzhou 221116, China, also with the Key Laboratory of Ecological Geology and Disaster Prevention, Ministry of Natural Resources, Xi'an 710054, China, and also with the College of Geological Engineering and Geomatics, Chang'an University, Xi'an 710054, China (e-mail: bqccumt@gmail.com).

Hanwen Yu is with the School of Resources and Environment, University of Electronic Science and Technology of China, Chengdu, Sichuan 611731, China (e-mail: yuhanwenxd@gmail.com).

Wen Xiao is with the National Engineering Research Center of Geographic Information System, China University of Geosciences, Wuhan 430074, China (e-mail: wen.xiao@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3649025

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

as a constraint, PL methods achieve temporally consistent phase information for DSs [17].

Several PL techniques have been introduced, including the phase triangulation algorithm (PTA) used in SqueeSAR [13], the covariance matrix eigenvalue decomposition method in component extraction and selection synthetic aperture radar (CAESAR) [21], the coherence matrix eigenvalue decomposition method in phase-decomposition-based PS-InSAR (PD-PSInSAR) [22], and the eigendecomposition-based maximum likelihood estimator of the interferometric phase (EMI) [23]. The differences among these PL methods mainly arise from their distinct weighting strategies. The optimized DS phases obtained through these methods can then be used as input for TS-InSAR analysis, enabling the estimation of surface deformation parameters.

With the increasing deployment of multipolarization synthetic aperture radar (SAR) satellites, the volume of multipolarization SAR data has grown substantially. Because ground objects interact differently with electromagnetic waves of various polarizations, data from different polarizations offer complementary insights into the reflective properties of these objects. This growing availability of multipolarization SAR data has opened new avenues for research in DS phase optimization. Early polarization-based phase optimization primarily targeted PS targets, with approaches such as the exhaustive search polarimetric optimization (ESPO) method [24], the suboptimum scattering mechanism (SOM) method [25], and the polarimetric Stanford method for PS (PolStaMPS) [26], all based on numerical optimization. Additionally, the coherency matrix decomposition-based polarimetric persistent scatterer interferometry (CMD-PolPSI) method was introduced to improve the efficiency of polarization optimization process [27].

As DS-InSAR techniques have advanced, numerous studies have incorporated multipolarization information into the PL process. Notable developments include a PL method based on adaptive scattering mechanism selection by Mullissa et al. [28]; the joint-polarization PL (JPPL) method and the polarization total power stacking-based EMI (TP-EMI) method proposed by Shen et al. [29], [30], respectively; and a multipolarization DS phase optimization approach based on Kronecker product decomposition by Wang et al. [31]. Further contributions include the phase-decomposition-based polarimetric coherence optimization (PD-PCO) method [32], a multipolarization maximum likelihood estimation-based PL (MLE-MPPL) method [33], the MPPE-CME method based on coherence matrix optimization [34], and the total power polarization optimization combined with nonlocal PL (TP-NL-PL) method [35].

By leveraging information from multiple polarizations, the multipolarization PL (MPPL) method incorporates more comprehensive object characteristics compared to single-polarization approaches. Consequently, MPPL methods achieve enhanced phase optimization for DS targets, surpassing the performance of conventional single-polarization PL techniques. Most existing PL methods optimize the DS phase under the assumption that the phase remains stationary within the estimation window [36]. However, mining areas often experience significant ground subsidence, leading to pronounced phase instability in interferograms. Applying these methods without addressing phase instability can result in substantial errors in phase estimation. Therefore, removing deformation fringes from interferograms is essential to improve the accuracy of PL-based deformation phase estimation. Previous approaches to estimating deformation fringes have relied on fringe frequency-domain estimation within local windows [37], [38], [39], [40] or on deformation field models [41], [42]. Frequency-domain methods struggle to capture nonlinear phase variations in local windows and are highly sensitive to noise [42]. Model-based approaches, such as those using probability integral models [43], require numerous geological and mining parameters, complicating implementation and limiting their applicability in mining areas.

Although the MPPL method enhances PL estimation accuracy compared to single-polarization methods, it remains affected by the sample Coherence matrix (SCM) estimation bias. Since PL methods are sensitive to SCM quality, most current MPPL approaches focus on leveraging multipolarization information while overlooking inherent SCM bias, thereby reducing phase estimation accuracy. For example, the TP-EMI method simply aggregates SCMs from different polarization channels based on total power theory, without adequately addressing coherence matrix estimation accuracy. Similarly, the MLE-MPPL method is derived under the assumption of a complex circular Gaussian distribution, but in practice, the polarimetric coherence and interferometric matrices are estimated by averaging across multiple acquisitions and polarization channels, rather than using the true matrices, which further compromises the accuracy of DS phase estimation.

To address the challenges outlined above, this article proposes a mining-oriented neural-polarimetric PL (MNP-PL) method that integrates deep neural networks with multipolarimetric information to enhance DS phase estimation accuracy in mining areas. The MNP-PL approach comprises two main components. First, to mitigate phase instability in mining areas, a deep neural network-based equivalent single reference phase estimation (DEPE) method is developed to estimate the deformed equivalent single reference (ESR) phase. This estimated phase is then used to remove deformation fringes from the original ESR interferograms, ensuring phase stability during PL processing. Second, a polarization and spatial feature fusion-based PL (PSFF-PL) method is introduced to achieve precise PL estimation. In PSFF-PL, both the polarization and spatial characteristics of DSs are utilized in SCM estimation, effectively suppressing phase noise and deviations in the sample coherence magnitude matrix (SCMM). The accuracy of the PL solution is further improved by employing an adaptive nonlinear shrinkage (ANS) strategy for SCM estimation. The primary advantage of the proposed method is that it does not require prior information to remove deformation fringes from interferograms and can be readily adapted to various mining environments. Additionally, by fusing DS polarization with spatial domain features, the method enhances SCM estimation accuracy, leading to further improvements in DS phase estimation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

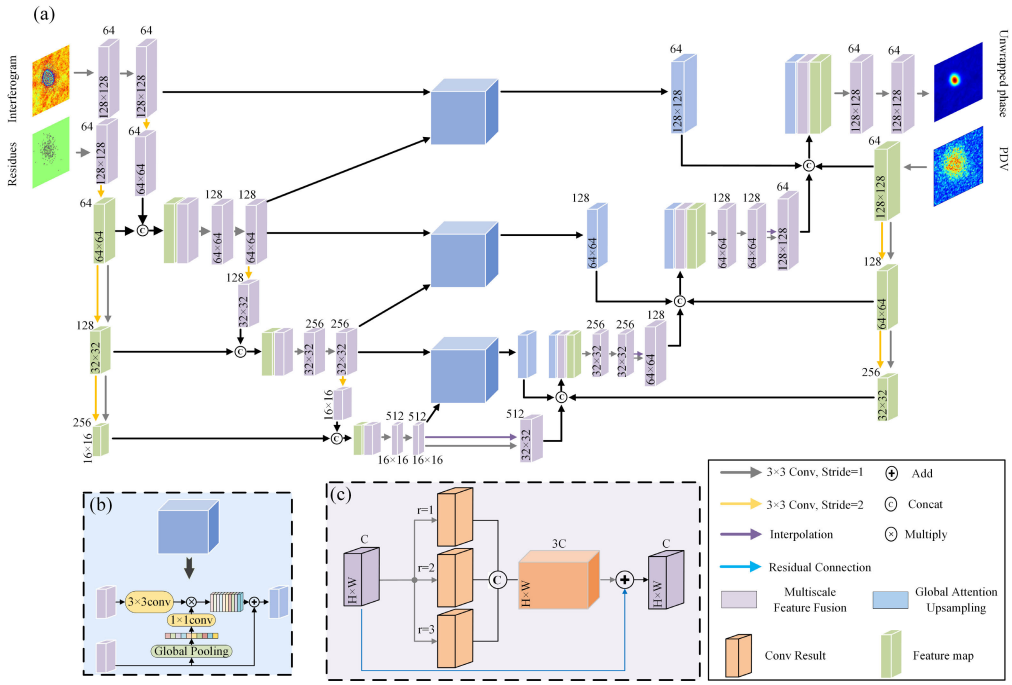



Fig. 1. Neural network architecture. (a) KLC-Net network framework, (b) global attention upsampling module, and (c) multiscale feature fusion module. Please see Chen et al. [46] for more information.

The remainder of this article is organized as follows. Section II describes the proposed MNP-PL algorithm, detailing the DEPE method, the PSFF-PL method, and the overall algorithm workflow. Section III presents the results and analysis of simulated experiments based on synthetic datasets. Section IV provides a quantitative evaluation and in-depth analysis of the proposed MNP-PL approach, comparing it with several established methods using real study areas and datasets. Section V discusses coherence estimation bias in the proposed and classical SCM construction methods based on simulation experiments and examines the impact of key parameter settings in the MNP-PL process on result accuracy. Finally, Section VI summarizes the main conclusions of this study.

## II. Method

### A. Deep Neural Network-Assisted ESR Phase Estimation

For a stack of fully polarimetric SAR (PolSAR) data, short temporal-baseline interferograms are first generated for the SLCs of different polarization channels on the basis of the temporal-baseline threshold. Afterward, the original short temporal-baseline interferogram phase is optimized based on the TP-ESM [44]. The optimized phase can be expressed as follows:

$$\varphi^{\text{TP-ESM}} = \frac{1}{2}\cdot\overline{|S_{\text{HH+VV}}|}^{2}\cdot e^{j\phi_{\text{HH+VV}}} + \frac{1}{2}\cdot\overline{|S_{\text{HH-VV}}|}^{2}\cdot e^{j\phi_{\text{HH-VV}}} + 2\cdot\overline{|S_{\text{VH}}|}^{2}\cdot e^{j\phi_{\text{VH}}} \quad (1)$$

where $\varphi^{\text{TP-ESM}}$ is the interferometric phase after TP-ESM optimization; $\phi_{\text{HH+VV}}$, $\phi_{\text{HH-VV}}$, and $\phi_{\text{VH}}$ are the interferometric phases corresponding to different polarization channels of the Pauli basis; and $\overline{|S_{\text{HH+VV}}|}$, $\overline{|S_{\text{HH-VV}}|}$, and $\overline{|S_{\text{VH}}|}$ are the time-series average amplitude information of the SAR images corresponding to the above polarization channels.

To obtain the ESR phase of the deformation region, this article introduces a deep learning network method, knowledge-and-deep-learning coupled network (KLC-Net), which couples domain knowledge with deep learning. Before the deformation phase is estimated using deep learning methods, the interferogram should be cropped according to the deformation range. To obtain the range of the deformation zone, the phase gradient stacking (PGS) method [45] was introduced to process the short temporal-baseline interferograms optimized by the TP-ESM. This PGS result is normalized, and pixels with normalized values greater than a certain threshold are selected as deformation pixels. Considering the spatial clustering characteristics of the deformation pixels in the mining area, the deformation boundary is determined on the basis of the positions of the screened pixels. Afterward, the interior of the boundary is filled to obtain the deformation zone clustering result.

Based on the clustering results of the deformation zones, the short temporal-baseline interferograms optimized by the TP-ESM method are cropped, and the phases of the deformation zones are extracted. Then, the extracted phase is unwrapped using KLC-Net. KLC-Net exhibits good model generalization performance and excellent noise resistance because it establishes a knowledge-learning network coupling framework under three-layer constraints of input variables, objective functions, and network structure. This framework effectively improves the ability and accuracy of InSAR phase disentanglement in mining areas [46]. Its network structure diagram is shown in Fig. 1. Please see Chen et al. [46] for more information.

The wrapped deformation phase is obtained by rewrapping the KLC-Net processing results within the deformation region. Based on the short temporal-baseline wrapped phase, a phase matrix $\Phi_{\text{stb}}$ is subsequently constructed for each pixel in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

deformation area, as defined in the following:

$$\Phi_{\text{stb}} = \begin{bmatrix} 1 & e^{j\varphi_{1,2}} & e^{j\varphi_{1,3}} & 0 & 0 & \cdots & 0 \\ e^{j\varphi_{2,1}} & 1 & e^{j\varphi_{2,3}} & e^{j\varphi_{2,4}} & 0 & \cdots & 0 \\ e^{j\varphi_{3,1}} & e^{j\varphi_{3,2}} & 1 & e^{j\varphi_{3,4}} & e^{j\varphi_{3,5}} & \cdots & 0 \\ 0 & e^{j\varphi_{4,2}} & e^{j\varphi_{4,3}} & 1 & e^{j\varphi_{4,5}} & \cdots & 0 \\ 0 & 0 & e^{j\varphi_{5,3}} & e^{j\varphi_{5,4}} & 1 & \cdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & e^{j\varphi_{n,m}} \\ 0 & 0 & 0 & 0 & \cdots & e^{j\varphi_{m,n}} & 1 \end{bmatrix} \tag{2}$$

where $\Phi_{\text{stb}}$ is the short temporal-baseline phase matrix and the element $\varphi_{m,n}$ is the interferometric phase between acquisitions $m$ and $n$ obtained by rewrapping the KLC-Net result. The short temporal-baseline phase matrix is solved using PTA as the constraint criterion. Then, the reference image phase is set to 0 to obtain the ESR phase of the deformation area. The solution process can be expressed as follows:

$$\hat{\varphi}_{\text{def}} = \arg\max_{\varphi} \left\{ \sum_{m=1}^{N} \sum_{n>m}^{N} \cos\left(\varphi_{m,n} - (\varphi_m - \varphi_n)\right) \right\} \tag{3}$$

where $\phi_m$ and $\phi_n$ are the $m$th and $n$th phases, respectively, of the deformation ESR phase solution results. Finally, the estimated ESR deformation phase is removed from the ESR interferogram phase $\phi_{\text{ori}}$, from which flat, topographic, and reference image phases have already been removed. The residual phase of the interferograms after fringe removal can be expressed as follows:

$$\varphi_{\text{res}} = \angle\left(e^{j\varphi_{\text{ori}}} \cdot \text{con}\, j\left(e^{j\varphi_{\text{def}}}\right)\right). \tag{4}$$

The residual phase $\phi_{\text{res}}$ contains only some unremoved deformation information and noise. The subsequent processing of the deformed area involves performing PL estimation on the residual phase.

Notably, during DEPE processing, it is necessary to extend the boundary of the deformation cluster by a radius equivalent to the maximum SCM estimation window size, thereby establishing a buffer zone. This measure ensures that, when constructing the SCM for pixels within deformation regions, both the reference pixel and its homogeneous counterparts utilize completely defringed phases. Conversely, for pixels located in nondeformation clusters, the SCM is directly constructed using the original ESR phases of the reference and homogeneous pixels. This strategy effectively mitigates phase discontinuities at the interfaces between deformation and nondeformation clusters in subsequent phase restoration procedures.

### *B. Polarization and Spatial Feature Fusion-Based PL*

For $N$ fully PolSAR data, the scattered signal vector can be expressed as follows [33], [47]:

$$\mathbf{x}_{\text{pol}} = [\mathbf{P_1}, \mathbf{P_2}, \mathbf{P_3}]^{\text{T}} \tag{5}$$

where $\mathbf{P}_i = [k_1^i, k_2^i, \ldots, k_N^i]^{\text{T}}$ and $k_N^i$ is the $i$th polarization channel signal corresponding to the $N$th SLC data. Based on a statistically homogeneous pixel (SHP) set $\Omega$ with a homogeneous sample number of $N_{\text{SHP}}$, the polarization interferometry sample covariance matrix $\hat{\mathbf{C}}_{\text{TSPol}} \in \mathbb{C}^{3N\times 3N}$ can be expressed as follows:

$$\hat{\mathbf{C}}_{\text{TSPol}} = \frac{1}{N_{\text{SHP}}} \sum_{\mathbf{y}_{\text{pol}}\in\boldsymbol{\Omega}} \mathbf{x}_{\text{pol}}\mathbf{x}_{\text{pol}}^{\text{H}} = \begin{bmatrix} \widehat{\mathbf{C}}_1 & \widehat{\boldsymbol{\Omega}}_{1,2} & \widehat{\boldsymbol{\Omega}}_{1,3} \\ \widehat{\boldsymbol{\Omega}}_{2,1} & \widehat{\mathbf{C}}_2 & \widehat{\boldsymbol{\Omega}}_{2,3} \\ \widehat{\boldsymbol{\Omega}}_{3,1} & \widehat{\boldsymbol{\Omega}}_{3,2} & \widehat{\mathbf{C}}_3 \end{bmatrix} \tag{6}$$

where $(\cdot)^{\text{H}}$ represents the conjugate transpose operation and $\widehat{\mathbf{C}}_i = 1/(N_{\text{SHP}}) \sum_{y_{\text{pol}}} K_i K_i^{\text{H}}$ is the sample covariance matrix from the $i$th polarimetric channel. The matrix structure is consistent with the single-polarization sample covariance matrix [34]. $\widehat{\boldsymbol{\Omega}}_{i,j} = 1/(N_{\text{SHP}}) \sum_{y_{\text{pol}}} K_i K_j^{\text{H}}$ is the cross covariance matrix between the $i$th and $j$th polarimetric channels. The covariance matrix $\widehat{\mathbf{C}}_i$ in the polarimetric–interferometric sample coherence matrix $\hat{\mathbf{C}}_{\text{TSPol}}$ represents the fully connected interferometric phase information of the $i$th polarization channel. After each sample covariance matrix $\widehat{\mathbf{C}}_i$ in the polarimetric–interferometric sample coherence matrix is normalized, the sample coherence matrix $\widehat{\mathbf{T}}_i$ of different polarization channels is obtained. Then, $\widehat{\mathbf{T}}_i$ is vectorized and expressed as follows:

$$\hat{\mathbf{T}}_i \to v_{\text{pol}}^i \in \mathbb{C}^{(N\times N)\times 1}. \tag{7}$$

The full polarization sample matrix $G_{\text{pol}}$ is constructed on the basis of the vectorized results of the sample coherence matrices of different polarization channels, where $G_{\text{pol}} = [v_{\text{pol}}^1, v_{\text{pol}}^2, v_{\text{pol}}^3] \in \mathbb{C}^{(N\times N)\times 3}$. The covariance matrix $C_{\text{pol}}$ of the polarization samples can be expressed as follows:

$$C_{\text{pol}} = \frac{1}{N^2}\left(G_{\text{pol}} - \bar{G}\right)^{\text{H}}\left(G_{\text{pol}} - \bar{G}\right). \tag{8}$$

In (8), $\bar{G} = \bar{v} \times I$, where $\bar{v}$ is the column mean of $G_{\text{pol}}, I \in \mathbb{R}^{1\times(n+1)}$ is a row vector, and all the elements in the vector are 1. The covariance matrix $C_{\text{pol}}$ is then subjected to eigendecomposition to obtain

$$C_{\text{pol}} = \sum_{i=1}^{3} \lambda_i \alpha_i \alpha_i^{\text{T}} = \lambda_1 \alpha_1 \alpha_1^{\text{T}} + \sum_{i>1}^{3} \lambda_i \alpha_i \alpha_i^{\text{T}} = C_{\text{DSM}} + C_{\text{others}}. \tag{9}$$

In (9), three eigenvalues $\lambda_1$–$\lambda_3$ can be obtained for fully polarimetric data, where $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3$. $C_{\text{DSM}}$ is the covariance matrix constructed from the signals associated with the dominant scattering mechanism (DSM) within the polarimetric sample matrix $G_{\text{pol}}$. $C_{\text{DSM}} = \lambda_1 \alpha_1 \alpha_1^{\text{T}}$, where the elements of the eigenvector $\alpha_1$ correspond to the characteristics of different polarimetric channel SCMs. The magnitude of each element represents the contribution weight of the corresponding polarimetric channel SCM feature to the SCM of the DSM signal. The eigenvector $\alpha_{\text{pol}}^1$ corresponding to the maximum eigenvalue $\lambda_1$ is adopted as the feature weighting vector. Multipolarization feature fusion (MPFF) is subsequently performed as follows:

$$\alpha_{\text{pol}} = \frac{\alpha_{\text{pol}}^1}{\sum \text{Re}\left(\alpha_{\text{pol}}^1\right)} \tag{10}$$

$$v_{\text{mpff}} = G_{\text{pol}} \times \alpha_{\text{pol}} \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

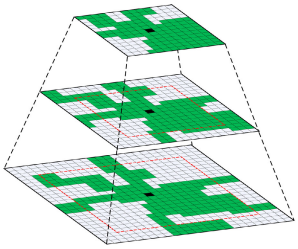


Fig. 2. Schematic of spatial pyramid construction. The base window size is 13 × 13, and the number of spatial pyramid layers is 3. The red dashed line indicates the position of the upper window within the lower window. The black pixels in the figure are reference pixels, and the green pixels are homogeneous pixels corresponding to the reference pixels.

where $\alpha_{\text{proj}}^{\text{pol}}$ is the normalized projection vector of $\alpha_1$ and $v_{\text{mpff}}^{(N\times N)\times 1}$ represents the sample coherence vector after MPFF. Upon obtaining the final vectorized SCM, dimensional restoration yields the multipolarization-fused SCM $\hat{\mathbf{T}}_{\text{mpff}} \in \mathbb{C}^{N\times N}$.

Traditional PL methods typically construct the SCM using homogeneous pixels within a fixed window size. While smaller windows better preserve the original phase characteristics, insufficient samples lead to significant phase noise in the SCM, introducing bias in the subsequent SCMM estimation. Conversely, larger windows effectively suppress interferometric phase noise and improve SCMM estimation accuracy but may cause oversmoothing errors. This is particularly problematic in regions with large deformation gradients, where spatial averaging in large multilook windows can distort dense phase fringes. To address this limitation, we propose a spatial pyramid feature fusion (SPFF) method for SCM construction, which can reduce the phase noise of the SCM while retaining the original effective phase features.

Before SCM estimation, an initial estimation window $\text{win}_1$ is defined, where $w_a$ and $w_r$ denote the window sizes in the azimuth and range directions, respectively. Starting from this initial window, new estimation windows are formed by outward expansion, incrementing each dimension by $(w_a-1)/2$ in the azimuth direction and $(w_{\text{r}}-1)/2$ in the range per expansion step. Assuming that $\text{win}_1$ expands outward $n-1$ times to yield $n$ total windows, the final size of $\text{win}_n$ becomes $[w_a+(w_a-1)/2^*(n-1),\ w_r+(w_r-1)/2^*(n-1)]$. All window sizes are constrained to odd numbers. If expansion yields an even size, the window is extended by one additional pixel. The spatial pyramid structure for an initial 13 × 13 window expanded twice is shown in Fig. 2. The initial window and its successive expansions collectively form the spatial pyramid. The initial and expanded windows form a spatial pyramid structure. Based on the window of each layer of the spatial pyramid, the polarimetric–interferometric sample coherence matrix is estimated, and the polarization features are fused to obtain the SCM $\hat{\mathbf{T}}_{\text{mpff}}$ of the fused polarization features in each window.

Assuming that a spatial pyramid structure with $s$ layers is constructed, the information of homogeneous scatterers at different spatial levels can be used according to (6)–(11) to obtain the sample coherence matrices of MPFF at different spatial levels. Like the polarization domain feature fusion, the $s$ sample coherence matrices at different spatial levels are vectorized, and then, the spatial sample group $G_{\text{spa}} = [v_{\text{spa}}^1, v_{\text{spa}}^2, \ldots, v_{\text{spa}}^s] \in \mathbb{C}^{(N\times N)\times s}$ is constructed. Afterward, its covariance matrix $C_{\text{spa}}$ is calculated, the eigenvector $\alpha_{\text{spa}}^1$ corresponding to the maximum eigenvalue of the covariance matrix is taken as the feature weight, and finally, feature fusion is performed on SCM samples from different layers of the spatial pyramid, which can be expressed as follows:

$$\alpha_{\text{spa}} = \frac{\alpha_{\text{spa}}^1}{\sum \text{Re}\left(\alpha_{\text{spa}}^1\right)} \tag{12}$$

$$v_{\text{spff}} = G_{\text{spa}} \times \alpha_{\text{spa}} \tag{13}$$

where $\alpha_{\text{spa}}$ is the normalized projection vector of $\alpha_1$ and $v_{\text{spff}}^{(N\times N)\times 1}$ represents the sample coherence vector after SPFF. After the final sample coherence vector is obtained, its dimension is restored, and finally, the SCM with the fusion of polarization and spatial features is obtained, that is, $\hat{\mathbf{T}}_{\text{psff}} \in \mathbb{C}^{N\times N}$.

To further improve the accuracy of the PL, ANS processing is performed on the SCM after feature fusion to ensure the accuracy of the inverse matrix of the SCMM [20]. The SCM after shrinkage can be expressed as $\hat{\mathbf{T}}_{\text{psff-ans}}$, where $\hat{\mathbf{T}}_{\text{psff-ans}} = \hat{\mathbf{S}}_{\text{psff-ans}} \circ \Phi_{\text{psff}}$ and $\Phi_{\text{psff}} = \exp(j\angle(\hat{\mathbf{T}}_{\text{psff}}))$. The main diagonal values of the SCMM after ANS are processed by the modified Cholesky method to make the matrix positive definite, and the nonmain diagonal elements of the SCMM after matrix shrinkage can be expressed as follows [20]:

$$s_{m,n} = 1 - \frac{1-\gamma_{m,n}^2}{1-\gamma_{\min}^2} \tag{14}$$

where $\gamma_{m,n}$ is the element in the $m$th row and $n$th column of the SCMM $|\hat{\mathbf{T}}_{\text{psff}}|$ and $\gamma_{\min}$ is the minimum value of all the elements in the SCMM $|\hat{\mathbf{T}}_{\text{psff}}|$.

In accordance with the EMI estimation framework, the final PL solution process can be expressed as follows:

$$\hat{\varphi}_{\text{PSFF-PL}} = \angle\left(\text{argmin}_{\mathbf{u}}\left\{\mathbf{u}^{\text{H}} \cdot \left(\left|\hat{\mathbf{T}}_{\text{psff-ans}}\right|^{-1} \circ \left(\hat{\mathbf{T}}_{\text{psff-ans}}\right)\right) \cdot \mathbf{u}\right\}\right). \tag{15}$$

The overall processing steps of the proposed PSFF-PL method are shown in Fig. 3. Finally, for the deformed region, the PL result of the residual phase is fused with the ESR phase estimated by the DEPE method to obtain the final phase optimization result of the entire interferograms, which is expressed as follows:

$$\hat{\varphi} = \angle\left(e^{j\varphi_{\text{res}}^{\text{pl}}} \cdot e^{j\varphi_{\text{def}}}\right) \tag{16}$$

where $\hat{\varphi}$ is the final optimized phase, $\varphi_{\text{res}}^{\text{pl}}$ is the PL optimization result of the residual phase, and $\varphi_{\text{def}}$ is the ESR deformation phase estimated by the DEPE method.

### C. MNP-PL Processing Chain

The overall processing steps of the proposed MNP-PL method are shown in Fig. 4. The processing flow of the MNP-PL method consists of six parts. The specific steps are as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

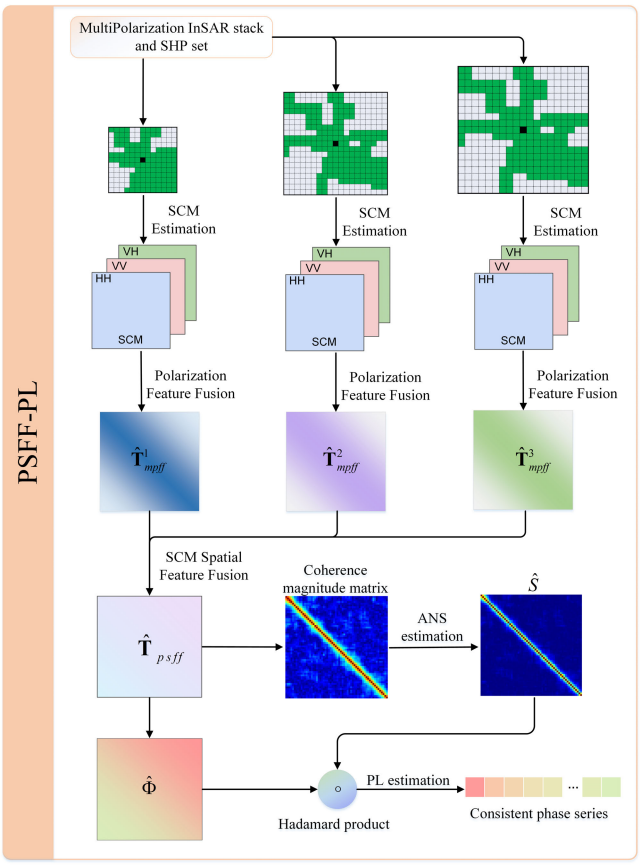



Fig. 3. Flowchart of the PSFF-PL method.

1) *Preprocessing:* First, the temporal polarization SAR (PolSAR) images are registered. Afterward, the polarization amplitude deviation of the original SLC data is optimized [24]. The optimized SLC intensity data are used to select homogeneous pixels using the Baumgartner–Weiss–Schindler test and dynamic interval estimation (BWS-DIE) method [12]. Finally, on the basis of the registered PolSAR data, the original ESR interferometric phase and short temporal-baseline interferograms less than a certain temporal-baseline threshold are generated.
2) *ESR Phase Defringing:* Based on the short temporal-baseline multipolarization interferograms obtained in step 1), the ESR phase of the deformation region is estimated using the DEPE method. Afterward, the deformation phase is removed from the ESR phase stack of each polarization channel to obtain the defringe multipolarization ESR phase stack.
3) *SCM Feature Fusion:* Based on the defringed multipolarization ESR phase stack obtained in step 2), an SCM that fuses polarization and spatial features is constructed based on homogeneous pixels.
4) *SCM Adaptive Nonlinear Shrinkage:* Based on the SCM obtained in step 3), the SCMM adaptively and nonlinearly shrunk.
5) *PL Estimation:* Based on the EMI estimation framework, the SCM is solved using (15) to obtain a consistent time-series phase.
6) *Phase Restoration in the Deformation Region:* The PL results of the residual phase in the deformation zone are combined with the ESR phase estimated by the DEPE method in step 2) to obtain the final phase optimization results for the deformation region.

## III. Simulation Experiment

### A. Simulation Data Generation and Parameter Configuration

To evaluate the performance differences of different PL methods in phase optimization, the experiment simulated $N = 30$ PolSAR acquisitions. The TS PolInSAR coherence matrix $\mathbf{C}_{\text{TSPolIn}}$ consists of the polarization coherence matrix $\mathbf{C}_{\text{pol}}$ and the interferometric complex coherence matrix $\mathbf{T}$ [48], which can be expressed as follows:

$$\mathbf{C}_{\text{TSPolIn}} = \mathbf{C}_{\text{pol}} \otimes \mathbf{T} \tag{17}$$

where $\otimes$ is the Kronecker product. Based on the extended Bragg scattering model, $\mathbf{C}_{\text{pol}}$ can be expressed as follows [49]:

$$\mathbf{C}_{\text{pol}} = \begin{bmatrix} \alpha_1 & \alpha_2 \sin c\,(2\beta) & 0 \\ \alpha_2^* \sin c\,(2\beta) & \alpha_3\,(1+\sin c\,(4\beta)) & 0 \\ 0 & 0 & \alpha_3\,(1-\sin c\,(4\beta)) \end{bmatrix} \tag{18}$$

where $^*$ denotes the conjugate operator, and $\alpha_1 = 1$, $\alpha_2 = 0.2 + 0.2i, \alpha_3 = 0.5$, and $\beta = 0.05\pi$. In (17), the elements in the matrix $\mathbf{T}$ can be expressed as $t_{m,n} = \gamma_{m,n} \exp(j(\varphi_{m,n}))$. According to the exponential decorrelation model [50], [51], $\gamma_{m,n}$ can be expressed as $\gamma_{m,n} = (\gamma_0 - \gamma_\infty) \exp(-|\Delta t|/\tau) + \gamma_\infty$, where $\gamma_0$ and $\gamma_\infty$ represent the short-term decay and long-term persistent coherence, respectively; $|\Delta t|$ represents the temporal baseline of the two images; and $\tau$ represents the temporal decorrelation constant. In this simulation experiment, the following parameters are set: $\gamma_0 = 1$, $\gamma_\infty = 0.05$, $|\Delta t| = 12$, and $\tau = 36$, and the radar wavelength is set to 5.6 cm. The first SLC was used as the reference image in the experiment to obtain the ESR phase. The interferogram size was $450 \times 600$. The intensity value of the true SLC stack was used to select homogeneous pixels to minimize the influence of homogeneous pixels on phase estimation. The true intensity map of the SLC and true ESR phase with a temporal baseline of 348 days is shown in Fig. 5(a) and (b), respectively.

### B. Comparison and Accuracy Analysis of the Reconstructed Phase

In the simulation experiments, EMI [23], TSO-PL [20], TP-EMI [30], MLE-MPPL [33], the proposed PSFF-PL method, and the MNP-PL method were compared and tested. In the simulation experiments, the fixed estimation window sizes of the other methods and the initial estimation window sizes of the PSFF-PL and MNP-PL methods are set to $13 \times 13$. As shown in the results presented in Fig. 6, the first to fourth columns present the optimization results of different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

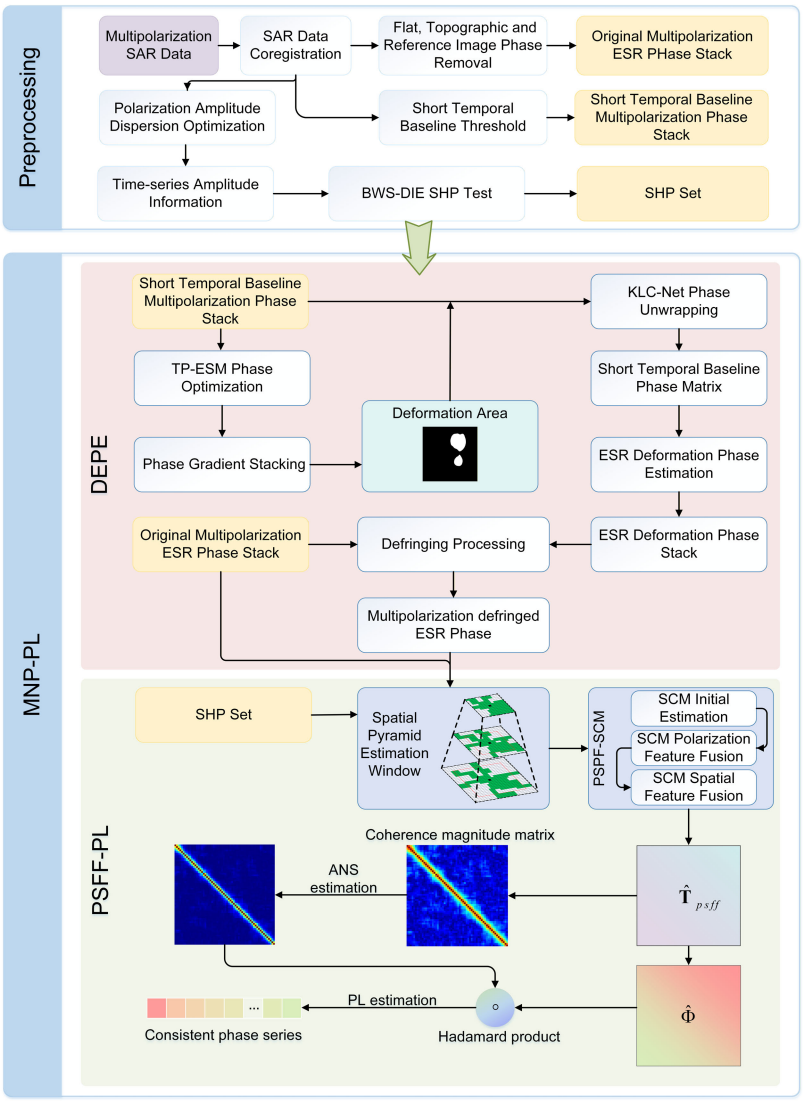



Fig. 4. Flowchart of the MNP-PL method.

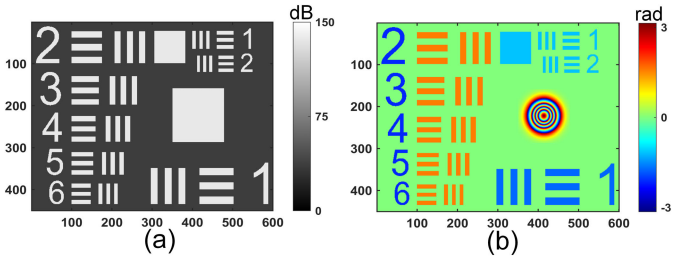



Fig. 5. Simulated intensity map and interferometric phase. (a) True intensity map. (b) Noise-free ESR phase with a temporal baseline of 348 days.

methods for temporal baselines of 36, 84, 168, and 348 days, respectively.

Fig. 6(a1)–(a4) shows the true phase under the four temporal baselines. The phase fringes in the deformation region gradually become denser as the temporal baseline increases. Fig. 6(b1)–(b4) shows the phase with decorrelated noise corresponding to the data in Fig. 6(a1)–(a4). The longer the temporal baseline, the higher the noise level of the interferogram. The PL estimation results of the EMI, TSO-PL, TP-EMI, and MLE-MPPL methods are relatively similar under two short temporal baselines (48 and 96 days) [Fig. 6 (c1), (c2), (d1), (d2), (e1), (e2), (f1), and (f2)], and the original fringe information can be retained in the deformed area; when the temporal baseline reaches 168 and 348 days, as the noise of the original interferogram increases, the noise of the corresponding optimization results also increases. Among the various methods, the TP-EMI method and MLE-MPPL method provide phase results with lower noise than the single-polarization EMI method [Fig. 6 (e3), (e4), (f3), and (f4)], which also indicates that multipolarization data increase PL accuracy. However, in the deformation region, the results estimated by the EMI, TP-EMI, and MLE-MPPL methods exhibit phase aliasing and fail to fully recover the original fringe features. Compared with the EMI, TP-EMI, and MLE-MPPL methods, the TSO-PL method is superior in suppressing noise over long temporal baselines and can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

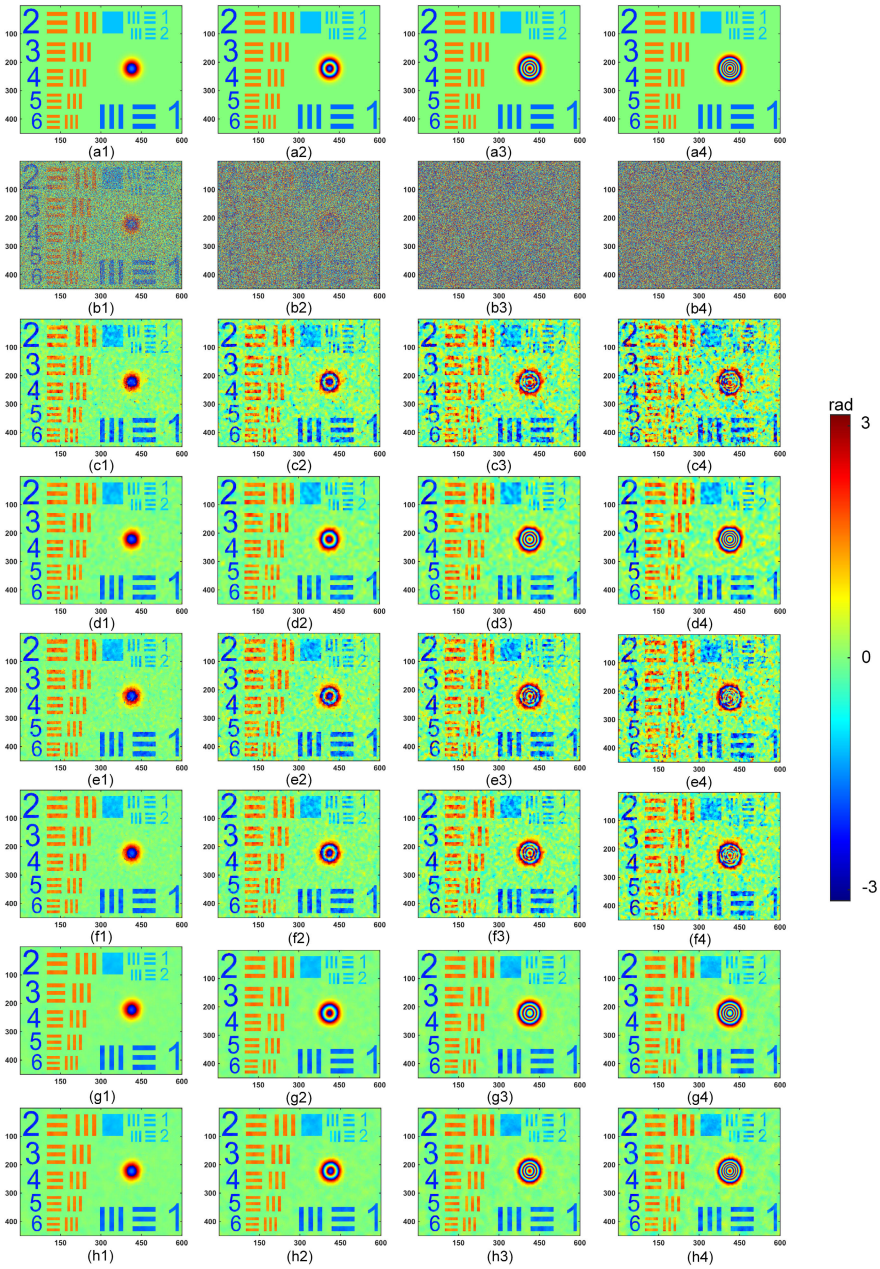



Fig. 6. Comparison of phase optimization results. The first to fourth columns correspond to the first to $i$th interferograms ($i$ = 4, 8, 15, and 30, corresponding to time intervals of 36, 84, 168, and 348 days, respectively) and the optimization results of the corresponding methods, where (a1)–(a4) represent the true phase, (b1)–(b4) are the noisy phases corresponding to (a1)–(a4), (c1)–(c4) are the single-polarization EMI phase optimization results, (d1)–(d4) are the TSO-PL phase optimization results, (e1)–(e4) are the TP-EMI phase optimization results, (f1)–(f4) are the MLE-MPPL phase optimization results, (g1)–(g4) are the PSFF-PL phase optimization results, and (h1)–(h4) are the MNP-PL (DEPE+PSFF-PL) phase optimization results.

recover relatively clear deformation fringes, but deviations still exist compared with the true deformation fringes [Fig. 6(d3) and (d4)]. The PSFF-PL method and MNP-PL estimation results indicate that both methods exhibit good noise reduction effects under the four temporal baselines. Additionally, they can accurately recover the interferogram fringes under long temporal baselines [Fig. 6(g3), (g4), (h3), and (h4)]. A comparison of the PSFF-PL results with the MNP-PL optimized results reveals that under short temporal baselines, the phase gradient in the deformation region is relatively small, and the optimized results from both methods are largely consistent [Fig. 6(g1), (g2), (h1), and (h2)]. However, Fig. 6(f4) and (g4) shows that the phase estimation results at the center of the deformation region exhibit phase inconsistencies, with PSFF-PL yielding significantly fewer fringes than MNP-PL does. This is because the original SLC data undergo spatial multilook averaging during PL estimation, which smooths out some deformation information and causes deformation features to be lost. In contrast, the MNP-PL method yields estimation results closer to the original true phase because it employs fringe removal processing.

To quantitatively evaluate the accuracy of the PL estimation results of the various methods, the root-mean-square error (RMSE) of each method was calculated (Fig. 7). Under

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

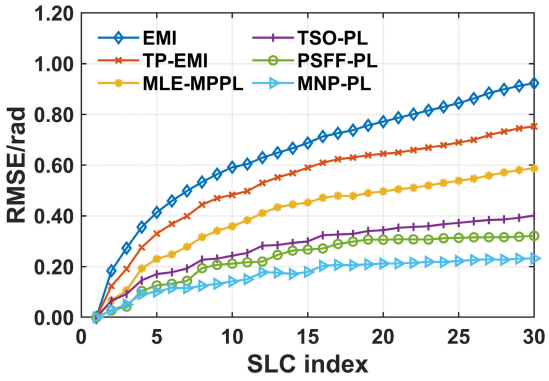



Fig. 7. RMSEs of the estimation results of different PL methods.

the simulated data in this experiment, the average time-series RMSEs for single-polarization EMI, TSO-PL, TP-EMI, MLE-MPPL, and MNP-PL were 0.6457, 0.2792, 0.5322, 0.4043, and 0.1676 rad, respectively. The average RMSE of the MNP-PL method improved the accuracy of the EMI, TSO-PL, TP-EMI, and MLE-MPPL methods by 74.04%, 39.97%, 68.50%, and 58.55%, respectively. Additionally, the computational time for the PL solution stage was measured for the EMI, TSO-PL, TP-EMI, MLE-MPPL, and proposed MNP-PL methods. The results indicate that the single-polarization EMI method takes 116.83 s, the TSO-PL method takes 1906.46 s, the TP-EMI method takes 216.72 s, the MLE-MPPL method takes 2327.74 s, and the proposed PSFF-PL method takes 408.96 s. In summary, although the proposed MNP-PL method has an intermediate computational time, it achieves the highest estimation accuracy among the comparison methods. Therefore, this method strikes a good balance between efficiency and accuracy, making it particularly valuable for monitoring deformations in mining areas.

To investigate the effects of different components of the proposed method on the final result accuracy, the single-polarization EMI method, the SCM MPFF-SCM method, the SCM polarization feature and SPFF (MPSPFF-SCM) method, the PSFF-PL method based on MPSPFF-SCM with SCM ANS, and the MNP-PL method integrated with defringe processing are compared. The results are shown in Fig. 8. Fig. 8(a1)–(a4) shows noise-free interferograms with corresponding temporal baselines of 36, 84, 168, and 348 days, respectively. Fig. 8 (b1)–(b4) presents the noise-added phases of Fig. 8(a1)–(a4), respectively. Compared with the estimation results of the single-polarization EMI method in Fig. 8(c1)–(c4), the results of the MPFF-SCM method in Fig. 8(d1)–(d4) significantly reduce the noise in the interferograms. Compared with the results obtained using the SCM polarization feature fusion method alone, the results obtained using the MPSPFF-SCM method [Fig. 8(e1)–(e4)] further suppress noise, and the fringe details at the long-term baseline subsidence are further preserved. However, fringe breaks still exist [Fig. 8(e3) and (e4)]. After applying the PSFF-PL method, which uses SCM adaptive nonlinear contraction based on the MPSPFF-SCM method, the phase optimization results show that the PL estimates in the deformation region have clear deformation stripe textures and essentially no phase aliasing [Fig. 8(f3) and (f4)]. However, compared with the number of phase fringes in true interferograms, the number of phase fringes is reduced. When the proposed MNP-PL method is used (in contrast to the PSFF-PL method, which adds defringe processing), the phase stationarity in the PL estimation window is guaranteed, which reduces the difficulty of the PL solution. As shown in Fig. 8(g1)–(g4), the MNP-PL method can better preserve the original deformation characteristics. Fig. 9 shows that compared with the single-polarization EMI method, the accuracies of the MPFF-SCM, MPSPFF-SCM, PSFF-PL, and MNP-PL methods improved by 36.32%, 57.22%, 63.81%, and 74.04%, respectively. The above experimental results verify the high accuracy and robustness of the proposed methods.

## IV. Case Study

### A. Study Area and Experimental Data

To validate the effectiveness of the proposed method, the Shilawusu Mining Area in the Inner Mongolia Autonomous Region of China was selected to test the algorithm's performance. The location characteristics of the study area are shown in Fig. 10. The surface of the study area is primarily composed of wind-deposited sand and vegetation, which aligns with the characteristics of DSs. InSAR phase quality is generally low in this region. The working panel is located in the central–southern part of the mine, and the mining direction is from south to north. The working panel has a width of 300 m and a length of 1135 m, and the mining period ran from September 16, 2017, to October 25, 2019. There were no mining activities from April 2018 to September 2019 because of a shutdown. The average mining depth was 655 m, with an average mining thickness of 9.87 m and an average coal seam dip angle of 2°. Surface deformation patterns were obtained using surface level observation stations (red dots in Fig. 10) with 50 measurement points (Z1–Z50) along the strike line, which were spaced 28 m apart on average.

In the case study, a total of 24 Sentinel-1 A ascending dual-polarization SAR images from September 8, 2017, to June 23, 2018, were selected as the experimental data. Considering the influence of temporal decorrelation, the image from January 6, 2018, was selected as the reference main image in the experiment. All the other images were registered with this reference image. The image spatiotemporal-baseline information is shown in Fig. 11(a). The polarization amplitude deviation optimization method [24] was used to optimize the registered dual-polarization SLC image. Afterward, the BWS-DIE method [46] was used to select homogeneous pixels. Finally, SRTM 1 ($30 \times 30$ m) data were used to remove the terrain phase in the SLC. Fig. 11(b) shows the RGB image synthesized from different polarization channel information of the experimental area, where $R = |\overline{VV}|, G = 4|\overline{VH}|$, and $B = |\overline{VV}|/4|\overline{VH}|$.

### B. Analysis of the Phase Optimization Results

In the experiments, the single-polarization EMI, TP-EMI, MLE-MPPL, TSO-PL, PSFF-PL, and MNP-PL methods were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

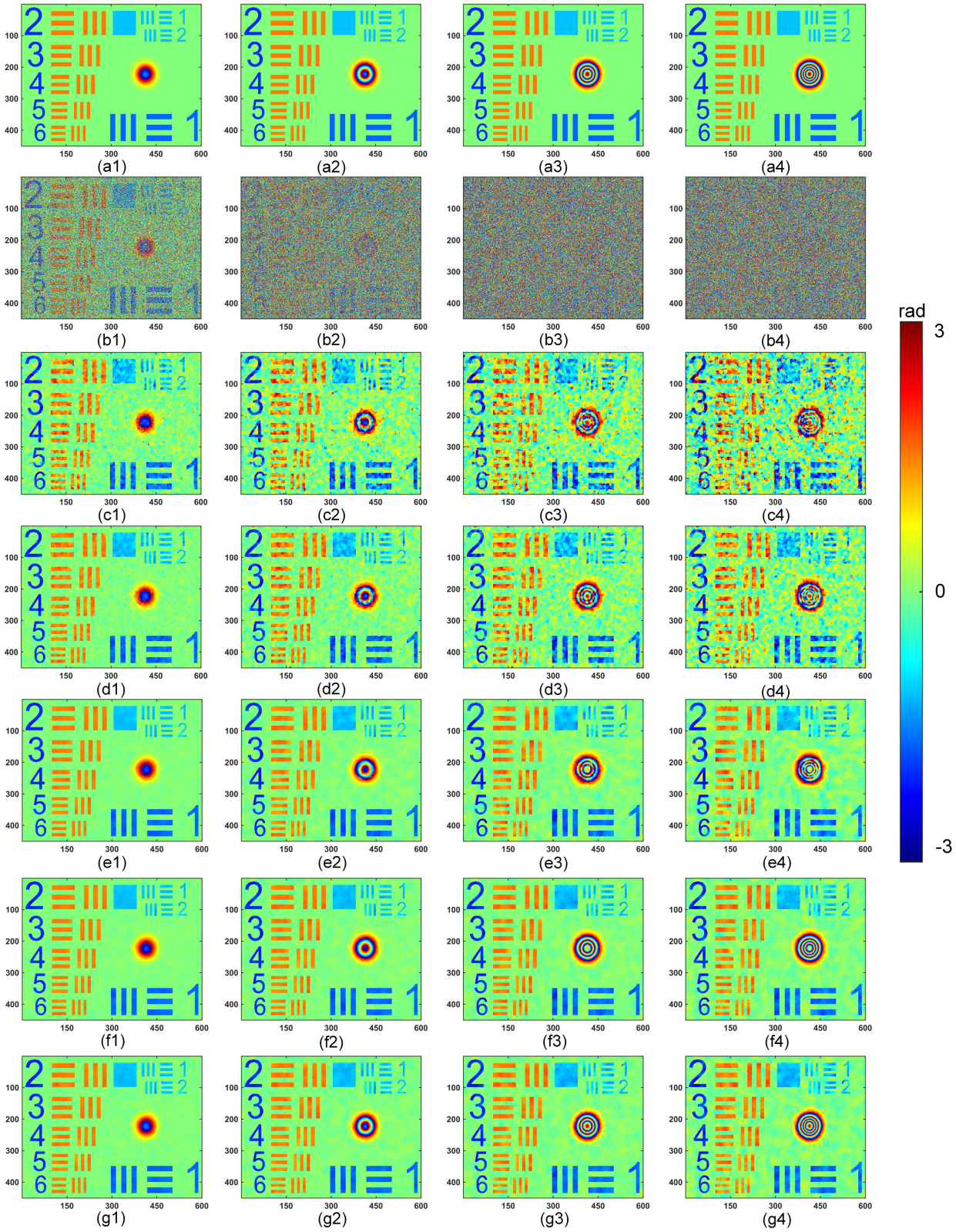



Fig. 8. Comparison of 2-D simulation experimental results. The first to fourth columns show the interferograms of the first and $i$th images ($i$ = 4, 8, 15, and 30) and the optimization results of the different methods, respectively, where (a1)–(a4) are the true phase, (b1)–(b4) are the noisy phase of the VV polarization channel, (c1)–(c4) are the single-polarization EMI phase optimization results, (d1)–(d4) are the MPFF-SCM phase optimization results, (e1)–(e4) are the MPSPFF-SCM phase optimization results, (f1)–(f4) are the PSFF-PL phase optimization results, and (g1)–(g4) are the MNP-PL phase optimization results.

selected for comparative analysis. The initial estimation window sizes for PSFF-PL and MNP-PL were set to 11 $\times$ 45 (range $\times$ azimuth), which is consistent with the fixed estimation window sizes of the other methods. To compare the performance of different PL methods, interferograms from three different temporal baselines (48, 108, and 168 days) were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

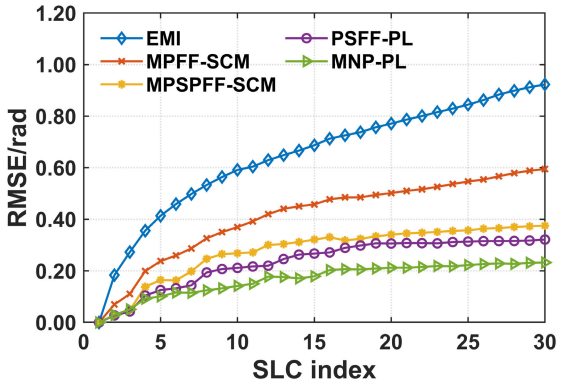



Fig. 9. RMSEs of the estimation results of different components of the proposed method.

selected as examples for analysis. The longer the temporal baseline, the stronger the phase noise in the images.

Fig. 12(a1)–(a7) shows the original phase with a temporal baseline of 48 days and the estimation results of the different PL methods. Under this temporal-baseline condition, the noise level of the original interferogram is relatively low and the interferometric fringes are relatively sparse. As shown in Fig. 12(a2)–(a4), under shorter temporal-baseline scenarios, the estimation results of EMI, TP-EMI, and MLE-MPPL are similar. Compared with EMI, TP-EMI, and MLE-MPPL, TSO-PL and the proposed PSFF-PL methods achieve further suppression of phase noise in their estimation results [Fig. 12(a5)–(a6)]. A comparison of Fig. 12(a7) and (a6) reveals that compared with the PSFF-PL method, the MNP-PL method achieves improved phase quality in both the deformation center region and edge regions, with deformation fringes that are more continuous and free of phase discontinuities.

Fig. 12(b1)–(b7) shows the original phase with a temporal baseline of 108 days and the estimated results of the different PL methods. Under this temporal-baseline condition, the original interferogram has a high noise level and features densely packed interferometric fringes. In such cases, the interferograms estimated using the single-polarization EMI, TP-EMI, and MLE-MPPL methods all exhibit significant phase aliasing at densely packed fringe regions [Fig. 12(b2)–(b4)]. Compared with the results of the single-polarization EMI method [Fig. 12(b2)], the TP-EMI and MLE-MPPL methods yield large numbers of random signals in the left half of the interferogram [Fig. 12(b3) and (b4)]. In contrast, the proposed PSFF-PL method yields relatively clear results [Fig. 12(b5)], significantly reducing phase aliasing and effectively restoring the original deformation features. Additionally, PSFF-PL achieves phase optimization effects similar to those of TSO-PL. However, both TSO-PL and PSFF-PL still exhibit minor discontinuities in the interferometric fringes within the deformation region [Fig. 12(b5) and (b6)]. Compared with TSO-PL and PSFF-PL, the MNP-PL method [Fig. 12(b7)] achieves superior recovery of high-quality, clear, and continuous deformation fringes in this high-noise scenario.

Fig. 12(c1)–(c7) shows the original phase with a temporal baseline of 168 days and the estimated results of the different PL methods. Under this temporal-baseline condition, the original interferogram has the highest noise level. As shown in Fig. 12(c1), the original interferogram contains virtually no effective phase information. However, owing to the phase recovery capability of PL algorithms in the time domain, all PL methods can significantly restore the original interferometric phase. However, owing to noise interference, as shown in Fig. 12(c2) and (c4), the estimation results of the single-polarization EMI, TP-EMI, and MLE-MPPL exhibit poor capability for restoring deformation fringes. As shown in Fig. 12(c5) and (c6), compared with the previous PL methods, the TSO-PL and PSFF-PL methods significantly improve the recovery capability of deformation fringes; however, discontinuity occurs in the interferometric fringes at the center of the deformation region. Compared with the TSO-PL and PSFF-PL methods, the MNP-PL method [Fig. 12(c7)] still demonstrates optimal performance under the longest temporal-baseline scenario, effectively restoring the actual deformation fringes.

To more intuitively illustrate the impact of the nonstationary deformation phase on the accuracy of the PL method in the interferogram (Fig. 13), enlarged views of the original VV polarization interferogram, the PSFF-PL method results, the ESR deformation phase obtained by deep neural network-assisted processing, the PSFF-PL results of the residual phase after fringe removal, and the final MNP-PL results corresponding to regions A and B in Fig. 12 (b7) and areas C and D in Fig. 12 (c7) are shown. The original interferometric phases are shown in Fig. 13(a1)–(d1). The results in Fig. 13(a2)–(d2) reveal that minor interferometric fringe discontinuities still exist in the PSFF-PL results. After fringe removal of the original interferogram and estimation using the PSFF-PL method, the optimized results of the corresponding residual phases are shown in Fig. 13(a4)–(d4). As shown in the figures, owing to the removal of deformation fringes, the residual deformation phase gradient decreases, and the phase stability within the estimation window is satisfactory during PL estimation. Therefore, the adverse effects of dense deformation fringes on PL estimation are mitigated to a certain extent, further improving the accuracy of PL estimation. As shown in Fig. 13(a5) and (d5), the final MNP-PL phase estimation results display clear, continuous interferometric fringes with high phase quality.

In the experiments, the phase matrices of the 80th row and 1020th column of the image were used as examples to compare and analyze the phase matrix after feature fusion and the single-polarization phase matrix. The single-polarization VV phase matrix, VH polarization phase matrix, phase matrix with fused polarization features, and phase matrix with fused polarization and spatial features are shown in Fig. 14(a)–(d), respectively. A comparison of the regions marked by the black dashed lines in Fig. 14 reveals that compared with the two single-polarization phase matrices [Fig. 14(a) and (b)], the phase matrix that fuses multipolarization information [Fig. 14(c)] can eliminate the phase noise caused by non-primary scattering signals while reducing the phase jumps of some interferograms in the fully connected phase matrix. After polarization and spatial feature fusion, compared with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

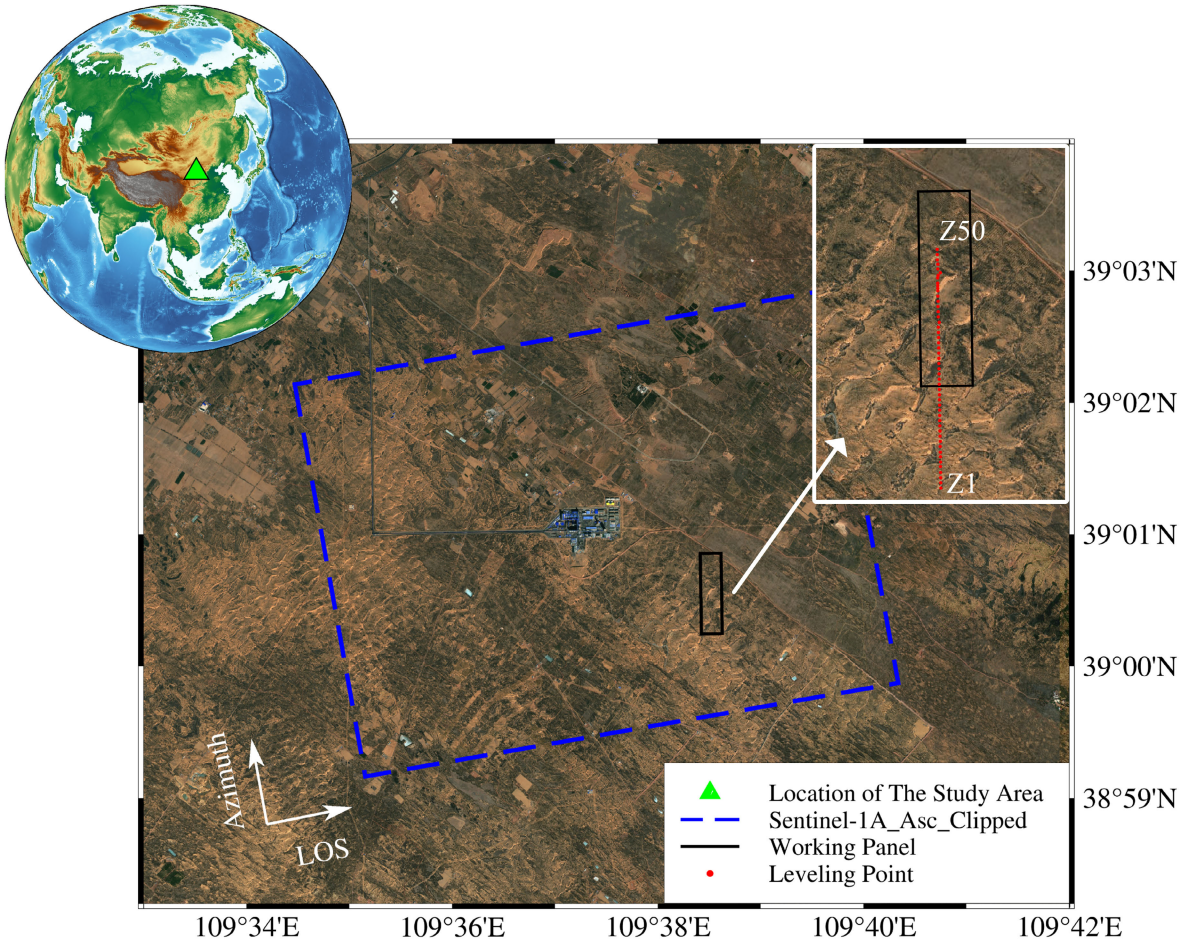



Fig. 10. Location of the study area. The green triangle indicates the study area, the blue dashed box indicates the clipped SAR image coverage, the black solid box indicates the working panel, and the red dots indicate the leveling stations. The leveling data were collected with an accuracy of ±5 mm/km round trip.

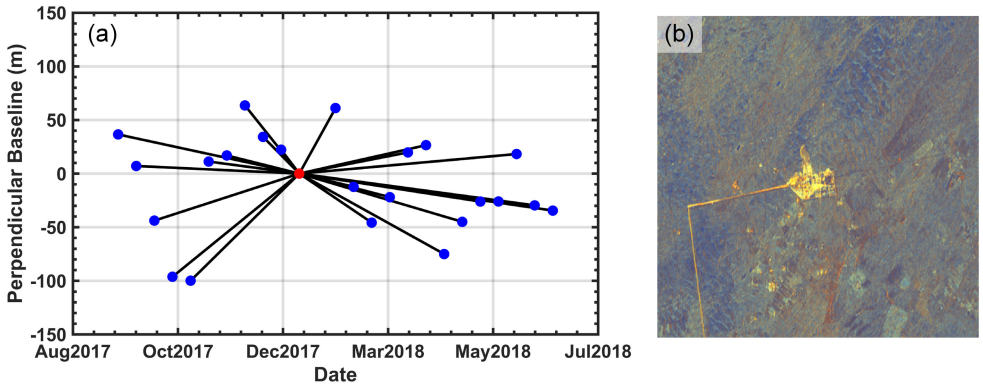



Fig. 11. Spatiotemporal-baseline distribution map and polarized RGB composite map of the experimental area. (a) Temporal and perpendicular baselines of the experimental data based on Sentinel-1 A images, where the red dot represents the reference image and the blue dots represent other images. (b) RGB image synthesized from different polarization channel information of the experimental area, where $R = |\overline{VV}|, G = 4|\overline{VH}|$, and $B = |\overline{VV}|/4|\overline{VH}|$.

the previous three-phase matrices, the SCM [Fig. 14(d)] further suppresses noise.

To quantitatively evaluate the phase estimation results of single-polarization EMI, TSO-PL, TP-EMI, MLE-MPPL, and the proposed MNP-PL algorithm, three evaluation indices—phase standard deviation (PSD) [52], absolute phase gradient sum (SPD) [53], and residual point number (RPN) [54]—are introduced for quantitative analysis. As shown in the data in Table I, after phase optimization using different PL methods, the three indicators of PSD, SPD, and RPN of all interferograms were significantly reduced, noise was significantly suppressed, and residual points were effectively removed. However, compared with the single-polarization EMI, TP-EMI, and MLE-MPPL methods, the proposed MNP-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

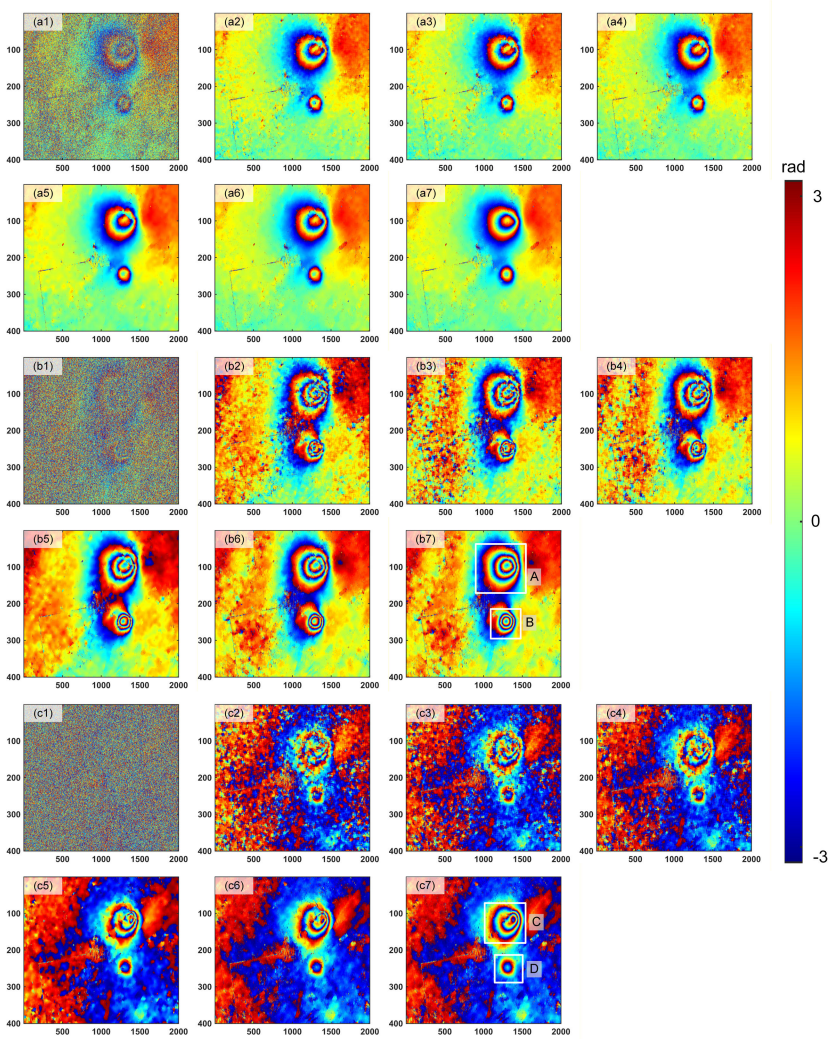



Fig. 12. Comparison of different PL method results under different temporal baselines. For each group of (a1)–(a7), (b1)–(b7), and (c1)–(c7), the seven panels are arranged in two rows (top row: 4; bottom row: 3). The top row consists of the raw input (original interferogram), followed by the results of EMI, TP-EMI, and MLE-MPPL. The bottom row shows the results of TSO-PL, PSFF-PL, and MNP-PL. Sample A (a-series) has a temporal baseline of 48 days, Sample B (b-series) 108 days, and Sample C (c-series) 168 days. Each group's first image is the original interferogram, whereas the remaining six images correspond to the six PL methods. Regions A, B, C, and D represent four typical areas. Regions A and B represent the deformation areas for the 108-day temporal baseline, whereas regions C and D represent the deformation areas for the 168-day temporal baseline. Due to the influence of the temporal baseline and mining progress, interference fringes are densely packed in regions A and B, and the original interferogram corresponding to regions C and D exhibits severe noise.

TABLE I
QUALITY EVALUATION RESULTS OF THE PHASE ESTIMATION WITH DIFFERENT METHODS

| | PSD/rad | SPD/rad | RPN/pixel |
|---|---|---|---|
| Original | 1.3603 | 1253653.80 | 330312 |
| EMI | 0.2462 | 209916.13 | 83860 |
| TSO-PL | 0.1022 | 86716.71 | 34821 |
| TP-EMI | 0.2319 | 197850.77 | 79866 |
| MLE-MPPL | 0.2192 | 182087.45 | 71484 |
| PSFF-PL | 0.1547 | 123952.34 | 49255 |
| MNP-PL | 0.1501 | 120477.13 | 48357 |

PL method performed better in terms of phase optimization, with lower values for the three evaluation indicators. Notably, owing to the excellent noise suppression capability of the homogeneous sample coherence matrix feature compression method in TSO-PL, the latter performs best among all the methods in terms of the three metrics. However, when the results in Fig. 12(b5) and (b7) are compared, the results reveal that although the TSO-PL method can provide smoother phases, it still lacks accuracy and completeness in restoring deformation zone fringes, with excessive smoothing occurring. In contrast, the MNP-PL method can produce clear and complete deformation stripes, further increasing the accuracy of PL estimation.

### C. Posterior Coherence Comparative Analysis

The experiment used posterior coherence as an evaluation index to evaluate the quality of the phase fitting results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

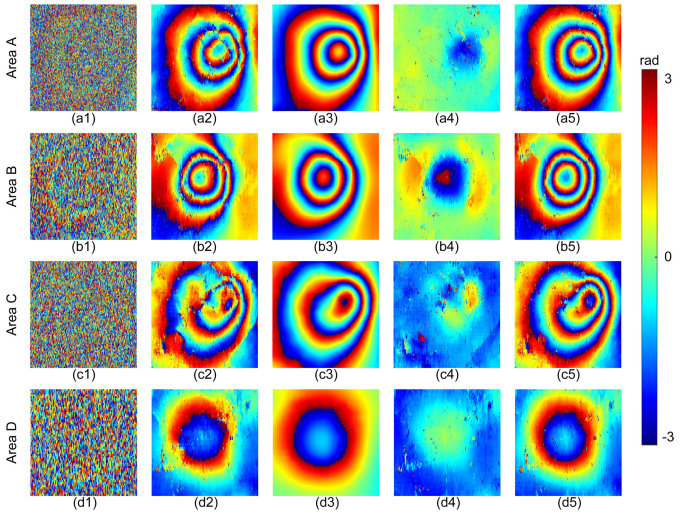



Fig. 13. PL results for different methods corresponding to regions A, B, C, and D. The first to fifth columns are the original single-polarized VV interferogram phase, PSFF-PL estimation results, deformation phase estimated by DEPE, PSFF-PL optimization results for the residual deformation phase, and MNP-PL optimization results, respectively. The temporal baseline of the interferograms in the first two rows is 108 days, whereas that of the interferograms in the last two rows is 168 days.

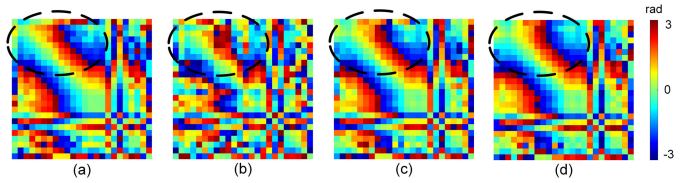



Fig. 14. Comparison of phase matrices for the pixel with row and column coordinates (80, 1020). (a) VV polarization phase matrix. (b) VH polarization phase matrix. (c) Phase matrix with fused polarization features. (d) Phase matrix with fused polarization and spatial features. The black dashed lines in the figure highlight examples in the phase matrices.

obtained by different PL methods. The formula for calculating posterior coherence is as follows [13]:

$$\gamma_{\mathrm{PL}} = \frac{2}{N(N-1)} \mathrm{Re}\left( \sum_{m=1}^{N} \sum_{n>m}^{N} \exp\left( j\left( \hat{\varphi}_{m,n} - \left( \hat{\theta}_m - \hat{\theta}_n \right) \right) \right) \right) \tag{19}$$

where $\hat{\varphi}_{m,n}$ is the interferometric phase for the $m$th row and $n$th column in the phase matrix after SCM optimization and $\hat{\theta}_m$ and $\hat{\theta}_n$ are the $m$th and $n$th phases of the pixel PL estimation results, respectively. Notably, for the MLE-MPPL method, the phase matrix is similar to that of the method used to obtain the true SCMM and is approximated by the estimated value of the SCM, where $\hat{\mathbf{T}} = 1/3 \sum_{i=1}^{3} \widehat{\mathbf{T}}_i$ in which $i$ is the number of polarization channels.

Fig. 15(a)–(f) shows the posterior coherence results for the six methods: EMI, TP-EMI, MLE-MPPL, TSO-PL, PSFF-PL, and MNP-PL, respectively. The posterior coherence distribution histograms for the results obtained from the six methods are shown in Fig. 16. As shown in Fig. 15(a)–(c), the posterior coherence results of the single-polarization EMI, TP-EMI, and MLE-MPPL methods are relatively consistent. Owing to its excellent noise reduction capability, the

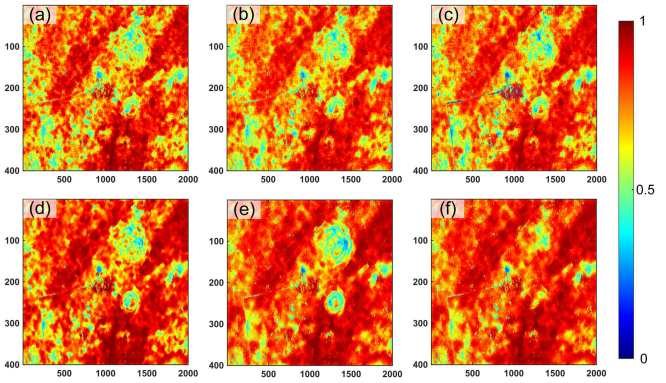



Fig. 15. Comparison of the posterior coherence of different methods. (a) Single-polarization EMI result, (b) TP-EMI result, (c) MLE-MPPL result, (d) TSO-PL result, (e) PSFF-PL result, and (f) MNP-PL method result.

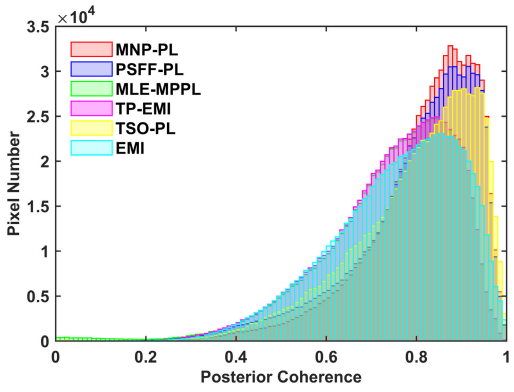



Fig. 16. Histograms of the posterior coherence corresponding to different PL methods.

TSO-PL method [Fig. 15(d)] shows a certain improvement in posterior coherence compared with the first three methods. Compared with the previous methods, the proposed PSFF-PL method [Fig. 15(e)] significantly increases the posterior coherence in nondeformed regions. However, the posterior coherence of the proposed PSFF-PL method in partially deformed areas is lower than that of the previous three methods. The reason is that the PL results of the first three methods exhibit phase aliasing in the deformation region, leading to pseudo-smoothing phenomena [Fig. 12(b2)–(b4)], thereby yielding artificially high posterior coherence [18]. The MNP-PL method uses fringe removal data for PL estimation. Since the assumption of phase stability in the estimation window is satisfied after the fringe phase is removed, the difficulty of the PL solution is reduced, thereby further improving the posterior coherence of the PL estimation. As shown in Fig. 15(f), compared with the other five methods, the MNP-PL method significantly increases posterior coherence in the deformation region.

### *D. Deformation Estimation Results and Accuracy Analysis*

To investigate the impact of different PL algorithms on the final deformation monitoring results, we selected pixels with a posterior coherence greater than 0.5 and more than 30 homogeneous pixels as DS points on the basis of the optimized phase results from five methods: EMI, TP-EMI, MLE-MPPL,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

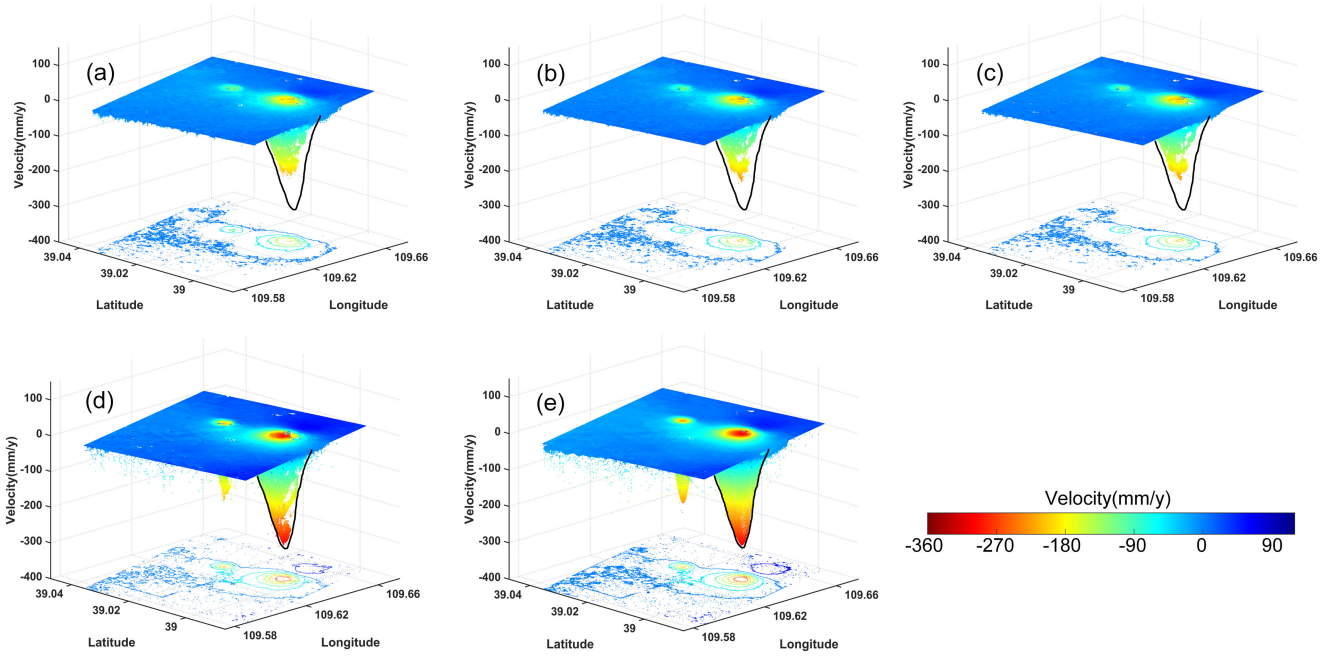



Fig. 17. Deformation rate results obtained using different PL methods. (a) EMI deformation results. (b) TP-EMI deformation results. (c) MLE-MPPL deformation results. (d) TSO-PL deformation results. (e) MNP-PL deformation results. Note that the upper layers in (a)–(e) are 2-D deformation rate maps for the various methods, where negative values represent subsidence and positive values represent uplift. Below the 2-D deformation rate maps are 3-D scatter plots of monitoring points with deformation rates less than 0, and at the bottom are contour line results corresponding to the deformation rate maps. The black curves in the figure represent the surface deformation rate range obtained by the MNP-PL method.

TSO-PL, and MNP-PL. These DS points were then fused with PS points with an amplitude deviation index of less than 0.4. The StaMPS method [10], [11] was subsequently used for the deformation solution. The deformation rate maps obtained by various methods are shown in Fig. 17(a)–(e).

As shown in Fig. 17(a)–(c), the first three methods underestimate the subsidence rate of the ground surface. Owing to the influence of deformation fringes on PL estimation, as shown in the posterior coherence results in Fig. 15, the EMI, TP-EMI, MLE-MPPL, and TSO-PL methods all have regions in the deformation zone where the posterior coherence of pixels is below the set threshold, leading to the discarding of some points in the deformation zone. As clearly shown in Fig. 17, the deformation rate results of the EMI, TP-EMI, MLE-MPPL, and TSO-PL methods differ within the deformation zone. In contrast, the proposed MNP-PL method maintains high posterior coherence within the deformation zone. Therefore, compared with the previous methods, the proposed method retains more valid monitoring points, and its monitoring results exhibit virtually no gaps within the deformation zone compared with the results of the previous methods.

To quantitatively evaluate the accuracy of the deformation results obtained using different methods, the deformation data from different PL methods were compared with the leveling observation station (Fig. 10) monitoring data. Surface monitoring data indicate that vertical subsidence is the primary deformation type in the study area. Therefore, the deformation data obtained from different PL methods in the LOS direction were directly projected onto the vertical direction and compared with the leveling data. The deformation data obtained from the surface leveling stations and the four PL methods

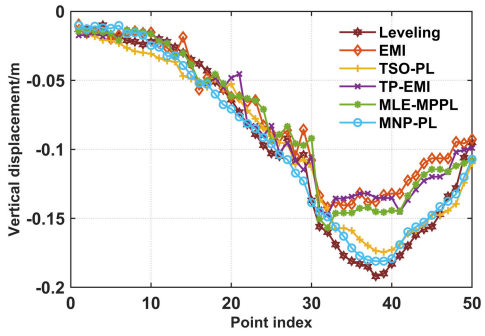



Fig. 18. Comparison of the deformation estimates of different PL methods with the field monitoring data. The point indices, numbered 1–50, correspond to the leveling measurement observations labeled Z1–Z50 in Fig. 10.

from September 8, 2017, to June 23, 2018, are compared in Fig. 18. Compared with the level data, the RMSE values of the deformation results obtained using the EMI, TP-EMI, MLE-MPPL, TSO-PL, and MNP-PL methods were 26.9, 23.5, 21.7, 11.9, and 8.6 mm, respectively. Compared with the EMI, TP-EMI, MLE-MPPL, and TSO-PL methods, the accuracy of the MNP-PL method improved by 68.02%, 63.40%, 60.36%, and 19.3%, respectively, demonstrating that the MNP-PL method offers higher accuracy and reliability.

## V. Discussion

### A. Comparison of the Accuracy of the SCMMs Estimated by Different Methods

The coherence magnitude of the SCM or a function with the coherence magnitude as the independent variable is usually used as the phase weight for PL estimation [55]. Therefore, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

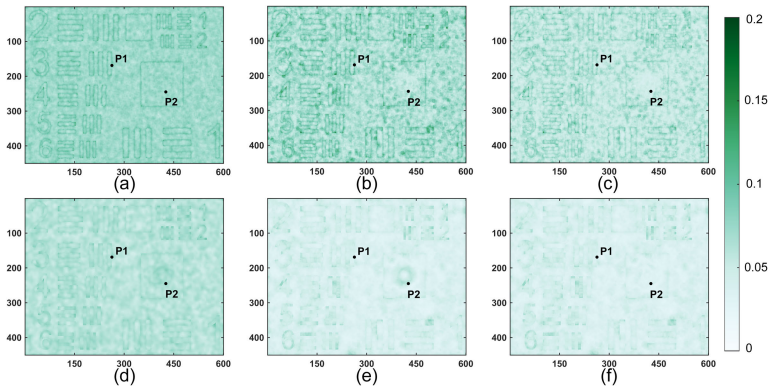



Fig. 19. Comparison of the SCMM deviations estimated by different PL methods. (a)–(f) SCMM deviations estimated by the EMI method, TP-EMI method, MLE-MPPL method, TSO-PL method, PSFF-PL method, and MNP-PL method, respectively. Points P1 and P2 represent characteristic points in the nondeformed and deformed areas, respectively.

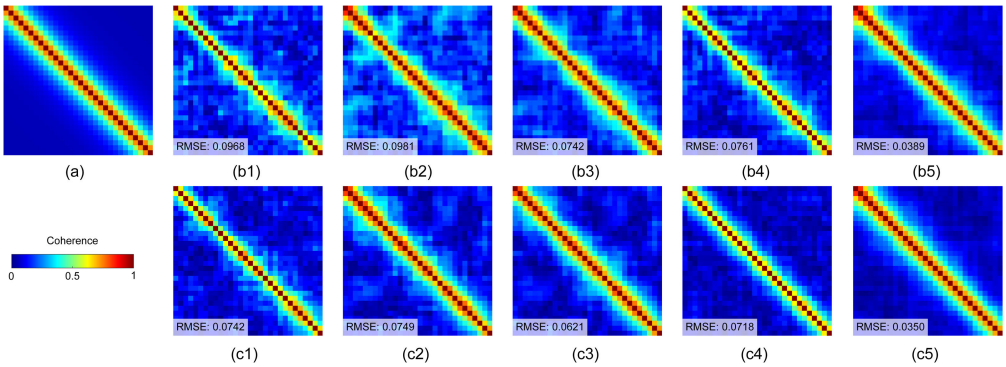



Fig. 20. Comparison of SCMM estimation errors between different PL methods in the nondeformed region and the deformed region. (a) Simulated true SCMM. (b1)–(b5) SCMMs constructed by different methods at point P1 in the nondeformed region: single-polarization EMI, the TP-EMI, MLE-MPPL, TSO-PL, and MNP-PL, respectively. (c1)–(c5) SCMMs constructed using different methods at point P2 in the deformed region: the single-polarization EMI, the TP-EMI, the MLE-MPPL, the TSO-PL, and MNP-PL, respectively. Since SCMMs are dimensionless, the corresponding RMSE values are also dimensionless.

accuracy of coherence estimation directly affects the accuracy of PL results. On the basis of the 2-D simulation experimental data in Section III, we explore the error between the SCMMs estimated by different PL methods and the true SCMM. The RMSEs of the SCMMs estimated by the single-polarization EMI method, TP-EMI method, MLE-MPPL method, TSO-PL method, proposed PSFF-PL method, and MNP-PL method are shown in Fig. 19. As shown in Fig. 19(a), the SCMM estimation error of the single-polarization EMI method is relatively large. In contrast, compared with the single-polarization EMI method, the TP-EMI method, MLE-MPPL method, TSO-PL method, proposed PSFF-PL method, and MNP-PL method all yield improved SCMM accuracy [Fig. 19(b)–(e)]. Compared with the previous four methods, the estimation accuracy of the SCMM is significantly greater with the PSFF-PL method since this approach integrates effective features from both the polarization and spatial domains when the SCM is estimated [Fig. 19(e)]. However, a comparison of the data in Fig. 19(e) and (f) reveals that the PSFF-PL and MNP-PL methods yield different SCMM estimation accuracies in the deformation area. This is because the MNP-PL method uses fringe removal data for PL estimation, which significantly reduces the coherence estimation deviation caused by deformation fringes.

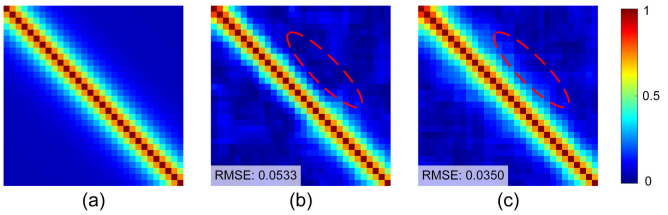



Fig. 21. Comparison of the SCMMs of the PSFF-PL and MNP-PL methods at point P2 in the deformation zone. (a) Simulated true SCMM, (b) SCMM estimated by the PSFF-PL method, and (c) SCMM estimated by the MNP-PL method. Notably, the PSFF-PL method underestimates the coherence within the area marked with red dashes. Since SCMMs are dimensionless, the corresponding RMSE values are also dimensionless.

Fig. 20 shows the SCMM estimated by different methods for points P1 in the nondeformed region and P2 in the deformed region in Fig. 19. It can be seen from Fig. 20 that the proposed MNP-PL method can estimate more accurate coherence magnitudes in both the deformed and nondeformed regions. Additionally, the proposed PSFF-PL method and MNP-PL method were compared in terms of estimating the SCMMs in deformation zone P2 (Fig. 21). A comparison of the true SCMMs in Fig. 21(a) reveals that the PSFF-PL method suffers from coherence underestimation because of the influence of the nonstationary deformation phase, as indicated by the region

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

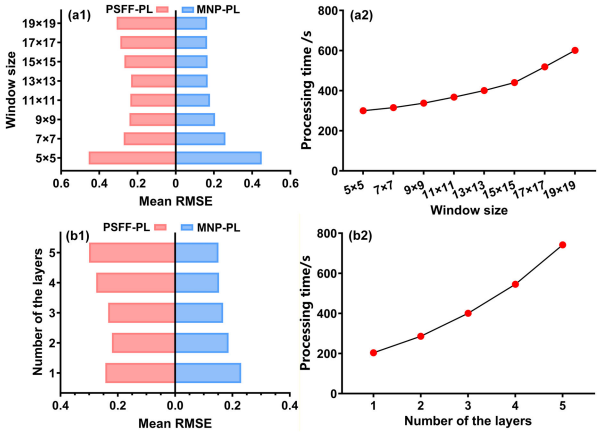



Fig. 22. Effect of different parameters on the accuracy of PL estimation and processing time. (a1) RMSE of phase estimation results using the PSFF-PL and MNP-PL methods under different initial estimation window sizes; (a2) effect of different initial estimation window sizes on the PL estimation time; (b1) RMSE of phase estimation results for the PSFF-PL and MNP-PL methods at different spatial pyramid layer numbers; and (b2) effect of different spatial pyramid layer numbers on the PL estimation time, where a layer number of 1 indicates that SCM SPFF processing is not used.

marked with red dashes in Fig. 21. The MNP-PL method produces a more accurate SCMM because fringe removal data are used to obtain the PL solution, which directly shows the impact of phase stability on the accuracy of the SCMM estimation results.

### *B. Effects of Different Parameters on the Accuracy of PSFF-Based PL Estimation*

The proposed PSFF-PL and MNP-PL methods require setting the initial estimation window size and the number of spatial pyramid layers for SCM feature fusion in the spatial domain. Based on the simulation experiment data from Section III, the following experiments were conducted: 1) the average RMSE of the PSFF-PL and MNP-PL methods with a fixed spatial pyramid layer count of 3 and an initial PL estimation window size ranging from $5 \times 5$ –$19 \times 19$ [Fig. 22(a1)] as well as the respective processing times required [Fig. 22(a2)] and 2) the average RMSE [Fig. 22(b1)] and the processing time required [Fig. 22(b2)] for the PSFF-PL and MNP-PL methods when the initial PL estimation window size is fixed at $13 \times 13$ and the spatial pyramid layer count varies from 1 to 5.

As shown in Fig. 22(a1), as the initial estimation window gradually increases, the accuracy of the SCM estimation is ensured because of the increase in the number of estimated samples. Moreover, the average RMSE of the PSFF-PL results first decreases but then increases because of the excessive size of the estimation window, leading to an increase in the SCM estimation bias. This is because the deformation phase is underestimated due to excessive smoothing of the phase in the deformation region. However, the MNP-PL method removes the ESR deformation phase before PL estimation, ensuring phase stability in the estimation window. Therefore, the fusion phase of the DEPE estimation results and the residual phase PL estimation results in the deformation zone yield higher accuracy than the PSFF-PL method does. Moreover, as the initial estimation window gradually increases, the average RMSE of the MNP-PL method gradually decreases and eventually converges. As shown in Fig. 20(a2), as the window gradually increases, the processing time also increases, and the overall trend of the algorithm's time consumption follows an exponential distribution.

The accuracy of the PSFF-PL method first decreases but then increases as the number of spatial pyramid layers increases [Fig. 20(b1)]. In contrast, the accuracy of the MNP-PL method gradually improves as the number of spatial pyramid layers increases. Compared with different estimation window sizes, the number of spatial pyramid layers has a greater effect on the time required for PL estimation when SCM feature fusion is used [Fig. 20(b2)]. Therefore, setting the number of spatial pyramid layers for SCM feature fusion to 2 or 3 is sufficient for general processing.

## VI. Conclusion

Mining areas are characterized by low coherence and significant surface subsidence, which often leads to substantial bias in phase estimation when using conventional PL methods. To address these challenges, this study introduces the MNP-PL method, specifically designed to enhance the accuracy of DS phase estimation in mining environments. Compared with existing approaches, the proposed method offers several key advantages: 1) the MNP-PL method effectively mitigates estimation bias caused by surface deformation in mining areas, resulting in more accurate phase estimation outcomes; 2) by integrating both polarization and spatial domain features during SCM construction, the method suppresses phase noise and reduces coherence estimation errors, leading to more reliable SCM estimation results; and 3) the proposed approach demonstrates strong capability in recovering deformation phases even under conditions of high fringe density and significant noise while maintaining a balance between estimation accuracy and computational efficiency.

The effectiveness of the MNP-PL method was evaluated using both simulation and real-world experiments. The simulation results demonstrate several key findings: 1) compared to the single-polarized EMI, TP-EMI, and MLE-MPPL methods, the MNP-PL method achieved the highest coherence estimation accuracy within deformation zones and 2) the MNP-PL method more accurately recovers phase information in both deformation and nondeformation regions while exhibiting superior noise reduction performance. Specifically, the phase estimation accuracy of MNP-PL improved by 74.04%, 39.97%, 68.50%, and 58.55% over the single-polarized EMI, TSO-PL, TP-EMI, and MLE-MPPL methods, respectively. The results from real-world case studies further highlight the advantages of the proposed approach.

1) The phase quality of the sample covariance matrix (SCM) constructed by the MNP-PL method is significantly enhanced compared to that of the single-polarization SCM. The DEPE-based MNP-PL approach restores clearer and more continuous deformation phase fringes, and, relative to the PSFF-PL method

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

without DEPE processing, substantially improves the posterior coherence of PL estimation in deformation areas.

2) The MNP-PL method delivers notable improvements in deformation monitoring performance and provides a more comprehensive picture of the deformation field in mining applications compared to single-polarization EMI, TP-EMI, and MLE-MPPL methods.
3) In terms of quantitative accuracy, the MNP-PL method achieves an RMSE of 8.6 mm when compared with leveling data, representing improvements of 68.02%, 63.40%, and 60.36% over the single-polarization EMI, TP-EMI, and MLE-MPPL methods, respectively.

The KLC-Net utilized in the MNP-PL method was primarily developed for horizontal coal seam mining. Its performance may be affected in areas characterized by inclined coal seam mining, and further validation is required for such scenarios. Additionally, the current approach employs a spatial pyramid structure to fuse SCM features at multiple spatial scales. Future research should explore optimal strategies for SCM feature fusion, tailored to the unique characteristics of different geological settings. Moreover, recent advances in deep learning present new opportunities for improving phase estimation. The integration of deep learning techniques with enhanced capabilities for spatial and temporal feature extraction holds promise for developing one-step PL estimation methods. Such advancements could enable direct phase estimation in low-coherence deformation regions with dense fringes, effectively mitigating the impact of phase instability on PL estimation accuracy.

## REFERENCES


[1] C. Wang, C. Zhu, X.-S. Wang, W. Tu, and Q. Li, "A new object-oriented SAR interferometry framework for monitoring urban deformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5227011.

[2] B. Chen et al., "Three-dimensional time-varying large surface displacements in coal exploiting areas revealed through integration of SAR pixel offset measurements and mining subsidence model," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111663.

[3] J. Cai et al., "Change detection of slow-moving landslide with multi-source SBAS-InSAR and light-U2Net," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104387.

[4] C. Yu et al., "Source characteristics and induced hazards of the 2025 M6.8 dingri earthquake, Xizang, China, revealed by imaging geodesy," *J. Earth Sci.*, vol. 36, no. 2, pp. 847–851, Apr. 2025.

[5] H. A. Zebker and J. Villasenor, "Decorrelation in interferometric radar echoes," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 5, pp. 950–959, Sep. 1992.

[6] R. F. Hanssen, *Radar Interferometry: Data Interpretation and Error Analysis*, vol. 2. Cham, Switzerland: Springer, 2001.

[7] A. Ferretti, C. Prati, and F. Rocca, "Nonlinear subsidence rate estimation using permanent scatterers in differential SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 5, pp. 2202–2212, Sep. 2000.

[8] A. Ferretti, C. Prati, and F. Rocca, "Permanent scatterers in SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 1, pp. 8–20, Jan. 2001.

[9] C. Werner, U. Wegmuller, T. Strozzi, and A. Wiesmann, "Interferometric point target analysis for deformation mapping," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2003, pp. 4362–4364.

[10] A. Hooper, H. Zebker, P. Segall, and B. Kampes, "A new method for measuring deformation on volcanoes and other natural terrains using InSAR persistent scatterers," *Geophys. Res. Lett.*, vol. 31, no. 23, p. 2004, Dec. 2004.

[11] A. Hooper, P. Segall, and H. Zebker, "Persistent scatterer interferometric synthetic aperture radar for crustal deformation analysis, with application to Volcán alcedo, Galápagos," *J. Geophys. Res., Solid Earth*, vol. 112, no. B7, p. 2006, Jul. 2007.

[12] B. Chen et al., "A new sequential homogeneous pixel selection algorithm for distributed scatterer InSAR," *GIScience Remote Sens.*, vol. 60, no. 1, Dec. 2023, Art. no. 2218261.

[13] A. Ferretti, A. Fumagalli, F. Novali, C. Prati, F. Rocca, and A. Rucci, "A new algorithm for processing interferometric data-stacks: SqueeSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 9, pp. 3460–3470, Sep. 2011.

[14] M. Jiang, X. Ding, and Z. Li, "Hybrid approach for unbiased coherence estimation for multitemporal InSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2459–2473, May 2014.

[15] H. Song et al., "An adaptive multilook approach for small sets of multitemporal SAR data based on adaptive joint data vector," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1161–1165, Jul. 2017.

[16] Y. Wang, Y. Deng, R. Wang, and J. Wang, "Adaptive multilooking based on complex patch for multitemporal interferometry," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 907–918, Mar. 2018.

[17] A. M. Guarnieri and S. Tebaldini, "On the exploitation of target statistics for SAR interferometry applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3436–3443, Nov. 2008.

[18] P. Shen, C. Wang, and B. An, "AdpPL: An adaptive phase linking-based distributed scatterer interferometry with emphasis on interferometric pair selection optimization and adaptive regularization," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113687.

[19] H. Liang, L. Zhang, X. Li, and J. Wu, "Coherence bias mitigation through regularized tapered coherence matrix for phase linking in decorrelated environments," *ISPRS J. Photogramm. Remote Sens.*, vol. 215, pp. 369–382, Sep. 2024.

[20] B. Chen et al., "TSO-PL: A novel phase linking method for DS InSAR based on a two-step strategy to optimize the sample coherence matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4416320.

[21] G. Fornaro, S. Verde, D. Reale, and A. Pauciullo, "CAESAR: An approach based on covariance matrix decomposition to improve multibaseline—Multitemporal interferometric SAR processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2050–2065, Apr. 2015.

[22] N. Cao, H. Lee, and H. C. Jung, "A phase-decomposition-based PSInSAR processing method," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 1074–1090, Feb. 2016.

[23] H. Ansari, F. De Zan, and R. Bamler, "Efficient phase estimation for interferogram stacks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4109–4125, Jul. 2018.

[24] V. D. Navarro-Sanchez, J. M. Lopez-Sanchez, and F. Vicente-Guijalba, "A contribution of polarimetry to satellite differential SAR interferometry: Increasing the number of pixel candidates," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 2, pp. 276–280, Apr. 2010.

[25] R. Iglesias, D. Monells, X. Fabregas, J. J. Mallorqui, A. Aguasca, and C. Lopez-Martinez, "Phase quality optimization in polarimetric differential SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2875–2888, May 2014.

[26] Z. Sadeghi, M. J. Valadan Zoej, A. Hooper, and J. M. Lopez-Sanchez, "A new polarimetric persistent scatterer interferometry method using temporal coherence optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6547–6555, Nov. 2018.

[27] F. Zhao and J. J. Mallorqui, "Coherency matrix decomposition-based polarimetric persistent scatterer interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7819–7831, Oct. 2019.

[28] A. G. Mullissa, D. Perissin, V. A. Tolpekin, and A. Stein, "Polarimetry-based distributed scatterer processing method for PSI applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3371–3382, Jun. 2018.

[29] P. Shen, C. Wang, C. Hu, J. Hu, H. Fu, and J. Zhu, "JPPL: A joint-polarization phase linking algorithm for phase optimization of TSPolInSAR data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102889.

[30] P. Shen, C. Wang, and J. Hu, "A polarization stacking method for optimizing time-series interferometric phases of distributed scatterers," *Remote Sens.*, vol. 14, no. 17, p. 4168, Aug. 2022.

[31] G. Wang et al., "A phase optimization method for DS-InSAR based on SKP decomposition from quad-polarized data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.

[32] Z. Zhang, X. Jiang, M. Wang, X. Liu, Q. Wang, and R. Li, "A phase-decomposition-based polarimetric coherence optimization method," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 110, Jun. 2022, Art. no. 102771.
[33] H. Xu, G. Zeng, W. Liu, and Y. Wang, "MLE-MPPL: A maximum likelihood estimator for multipolarimetric phase linking in MTInSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5202913.
[34] C. Zhao, H. Yu, and M. Jiang, "MPPE-CME: Multipolarimetric phase estimation for distributed scatterers with improved coherence matrix estimation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5213015.
[35] Y. Zhang et al., "Interferometric phase optimization based on total power polarization optimization and nonlocal phase linking," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5214515.
[36] Y. Wang and X. X. Zhu, "Robust estimators for multipass SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 968–980, Feb. 2016.
[37] H. A. Zebker and K. Chen, "Accurate estimation of correlation in InSAR observations," *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 2, pp. 124–127, Apr. 2005.
[38] G. Vasile et al., "High-resolution SAR interferometry: Estimation of local frequencies in the context of Alpine glaciers," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 4, pp. 1079–1090, Apr. 2008.
[39] S. Li, S. Zhang, Y. Gao, and Q. Zhang, "Slope-compensated interferogram filter with ESPRIT for adaptive frequency estimation," *IET Image Process.*, vol. 13, no. 8, pp. 1289–1294, Jun. 2019.
[40] B. Lei, L. Zhang, J. Wu, Z. Lu, and H. Liang, "Enhancing InSAR coherence estimation through local phase surface modeling," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[41] K. L. Feigl and C. H. Thurber, "A method for modelling radar interferograms without phase unwrapping: Application to the M5 fawnskin, California earthquake of 1992 December 4," *Geophys. J. Int.*, vol. 176, no. 2, pp. 491–504, Feb. 2009.
[42] X. Tian, X. Wu, H. Yu, and M. Jiang, "Positioning and mitigating mining-induced phase gradients for InSAR phase unwrapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11661–11669, 2025.
[43] J. Litwiniszyn, "Application of the equation of stochastic processes to mechanics of loose bodies," *Arch. Mech. Stos.*, vol. 8, no. 4, pp. 393–411, 1956.
[44] F. Zhao, L. Zhang, T. Wang, Y. Zhang, S. Yan, and Y. Wang, "An efficient polarimetric persistent scatterer interferometry algorithm for dual-pol Sentinel-1 data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3336–3352, 2023.
[45] D. Zhang, L. Zhang, J. Dong, Y. Wang, C. Yang, and M. Liao, "Improved phase gradient stacking for landslide detection," *Landslides*, vol. 21, no. 8, pp. 1829–1847, Aug. 2024.
[46] B. Chen et al., "A novel knowledge-learning coupling method for InSAR phase unwrapping of large surface displacements in coal mining areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4416320.
[47] Y. Chen, J. Hu, J. J. Mallorqui, H. Fu, W. Wu, and L. Zhang, "An interferometric phase optimization method joining polarimetric and temporal dimensions," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5208212.
[48] S. Tebaldini, "Algebraic synthesis of forest scenarios from multibaseline PolInSAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 12, pp. 4132–4142, Dec. 2009.
[49] S. Cloude, *Polarisation: Applications in Remote Sensing*. London, U.K.: Oxford Univ. Press, 2010.
[50] F. De Zan and P. Lopez-Dekker, "SAR image stacking for the exploitation of long-term coherent targets," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 502–506, May 2011.
[51] H. Ansari, F. De Zan, and A. Parizzi, "Study of systematic bias in measuring surface deformation with SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1285–1301, Feb. 2021.
[52] R. M. Goldstein and C. L. Werner, "Radar interferogram filtering for geophysical applications," *Geophys. Res. Lett.*, vol. 25, no. 21, pp. 4035–4038, Nov. 1998.
[53] Z. Li, W. Zou, X. Ding, Y. Chen, and G. Liu, "A quantitative measure for the quality of INSAR interferograms based on phase differences," *Photogramm. Eng. Remote Sens.*, vol. 70, no. 10, pp. 1131–1137, Oct. 2004.
[54] R. M. Goldstein, H. A. Zebker, and C. L. Werner, "Satellite radar interferometry: Two-dimensional phase unwrapping," *Radio Sci.*, vol. 23, no. 4, pp. 713–720, Jul. 1988.
[55] S. Li, S. Zhang, T. Li, Y. Gao, Q. Chen, and X. Zhang, "An adaptive phase optimization algorithm for distributed scatterer phase history retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3914–3926, 2021.



**Ningjie Liu** received the bachelor's degree from the College of Surveying and Geo-Informatics, North China University of Water Resources and Electric Power, Zhengzhou, China, in 2021, and the master's degree from Jiangsu Normal University, Xuzhou, China, in 2025.

His research focuses on interferometric synthetic aperture radar (InSAR) data processing and application.



**Bingqian Chen** (Member, IEEE) received the Ph.D. degree from China University of Mining and Technology, Xuzhou, China, in 2015.

He is an Associate Professor at the School of Geography, Geomatics and Planning, Jiangsu Normal University, Xuzhou. His research interests focus on radar data processing, mining subsidence monitoring and prediction, optimization algorithms, data fusion, and high-performance machine learning approaches for large-scale data analysis.



**Hanwen Yu** (Fellow, IEEE) is a Professor with the School of Resources and Environment, University of Electronic Science and Technology of China (UESTC). His research interests focus on interferometric synthetic aperture radar (InSAR) signal processing and its applications in Earth science.

Dr. Yu is the Editor-in-Chief of IEEE JOURNAL OF MINIATURIZATION FOR AIR AND SPACE SYSTEMS (JMASS), and currently serves as a member of the Editorial Board of *Proceedings of the IEEE* (P-IEEE). He is also a voting member of the IEEE Geoscience and Remote Sensing Society (GRSS) Administrative Committee (AdCom), serving during the 2023-2028 term.



**Wen Xiao** (Member, IEEE) received the B.S. degree in geodesy and geomatics from Wuhan University, Wuhan, China, in 2010, the M.S. degree (cum laude) in geoinformatics from the Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, Enschede, The Netherlands, in 2012, and the Ph.D. degree in geoinformation science and technology from the National Institute of Geographical Information and Forestry (IGN), Université Paris-EST, Paris, France, in 2016.

He has been a Lecturer and a University Research Fellow with Newcastle University, Newcastle upon Tyne, U.K. He is currently a Professor with China University of Geosciences, Wuhan, His research interests include 3-D mapping, laser scanning, photogrammetric computer vision, and their applications in smart cities and digital twins.

Dr. Xiao has been a Council Member of U.K. Remote Sensing and Photogrammetry Society (RSPSoc).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.



**Jiahui Liu** received the bachelor's degree in surveying and mapping engineering from Jiangsu Normal University, Xuzhou, China, in 2024, where she is currently pursuing the master's degree in cartography and geographic information engineering.

Her research interests include radar data processing and deformation monitoring.



**Junyun Wu** received the B.S. degree from the College of Information Engineering (College of Artificial Intelligence), Yangzhou University, Yangzhou, China, in 2024. He is currently pursuing the master's degree in electronic and information engineering with Jiangsu Normal University, Xuzhou, China.

His research interests include interferometric synthetic aperture radar (InSAR) data processing and multibaseline phase unwrapping.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:34:19 UTC from IEEE Xplore. Restrictions apply.